



Beyond Passive Explainability

Implementing and Evaluating a Human-in-the-Loop AI Recruitment System

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Des Weiteren ist an dieser Stelle natürlich auch meinen Eltern zu danken, die durch ihre allgegenwärtige Unterstützung (auch) das Schreiben dieser Arbeit ermöglicht haben. Dass Dank hier nicht auf die Unterstützung der Arbeit selbst oder das Studieren im Allgemeinen beschränkt ist, sei an dieser Stelle nur die der Vollständigkeit dienenden Ausformulierung des Selbstverständlichen – für nichtselbstverständliche Unterstützung.

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Kurzfassung

Die Landschaft der Human Resources (HR) wird zunehmend von Künstlicher Intelligenz (KI) geprägt, um den Bewerbungsprozess zu vereinfachen. Ein bekanntes Problem vieler KI-Modelle ist ihre Undurchsichtigkeit, auch bekannt als “Black Box”-Problem. Der Aufbau der Modelle verhindert eine transparente Einsicht in die Funktionsweise, damit also auch in die Motivation getroffener Entscheidungen. Ein Lösungsansatz, der versucht, dieses Problem zu verhindern, ist Explainable AI (XAI). Die durch XAI ermöglichten, mathematisch fundierten Einblicke werden in bestehenden Anwendungen allerdings nicht zu unterstützenden Tools weiterverarbeitet, sondern verbleiben als numerischer Output. So werden rein vermeintliche Einblicke geschaffen, die allerdings nur schwer verständlich sind und daher zu einer “Illusion von Fairness” führen.

Zur Lösung dieses Problems wird in dieser Arbeit ein neues, hybrides System vorgestellt und evaluiert. Ziel dieses Systems ist, Recruiter im Bewerbungsprozess zu unterstützen und ihnen zu ermöglichen, die Outputs des Systems aktiv zu hinterfragen und zu bestätigen. Ein Merkmal des Systems ist seine Interaktivität: Mithilfe von Slidern können Recruiter die Wichtigkeit verschiedener Hard- und Soft-Skills bestimmen. Large Language Models (LLMs) werden eingesetzt, um Daten aus Lebensläufen und einer Stellenausschreibung zu extrahieren, und um später Fragen für ein Bewerbungsgespräch und eine Zusammenfassung für alle Bewerbenden zu generieren. Das eigentliche Bewerten der Bewerbenden wird von einem Random Forest Classifier übernommen, der durch den Einsatz von TreeSHAP nachvollziehbare Einblicke ermöglicht.

Das System wurde abschließend durch eine Nutzerstudie ($N = 14$) evaluiert, bei der durch eine Kombination aus einem Think-Aloud Protocol, einem digitalen Fragebogen und einem semistrukturierten Interview eingesetzt wurde. Die quantitativen Ergebnisse des Fragebogens sprechen mit einem System Usability Score (SUS) von 88.93 für ein intuitives und verständlich aufgebautes Interface. Aus den qualitativen Ergebnissen geht hervor, dass die Interviewfragen und Zusammenfassung ein aktives Hinterfragen und Bestätigen der Outputs des Systems ermöglichen. Des Weiteren geht aus den Ergebnissen das Schüren von “Calibrated Trust” hervor. “Calibrated Trust” besteht, wenn die Balance zwei entgegengesetzter Verhaltensmuster, dem blinden Vertrauen in das System und der absoluten Ablehnung der Ergebnisse, eintritt: die Outputs werden akzeptiert, aber skeptisch hinterfragt. Die Arbeit schließt mit dem Vorstellen von Design Guidelines ab und trägt so zur Entwicklung zukünftiger XAI-Tools bei.

Abstract

Artificial Intelligence (AI) is increasingly utilized in Human Resource (HR) Management to streamline candidate screening. However, many high-performing models operate as opaque “black boxes”, prompting the adoption of Explainable AI (XAI) to ensure fairness and regulatory compliance. While current XAI methods, such as Shapley Additive Explanations (SHAP), provide mathematical transparency, they often suffer from low support for action: they produce complex numerical outputs that fail to actively support non-technical HR professionals in their subsequent decision-making and can thereby foster an “illusion of fairness”.

To address this, this thesis proposes and evaluates a novel hybrid Human-in-the-Loop framework that translates passive XAI outputs into actionable, human-centric insights. The architecture utilizes Large Language Models (LLMs) exclusively for deterministic resume parsing and natural language generation, while delegating the high-stakes task of candidate scoring to a mathematically auditable Random Forest classifier paired with TreeSHAP. Furthermore, an interactive recruiter dashboard enables a socio-technical “bargaining” process. This allows users to dynamically adjust features weights via Likert-scale sliders and instantly review updated scores, LLM-generated summaries and targeted interview questions based on the candidate’s exact feature attributions.

The framework’s practical viability was assessed through a mixed methods user study ($N = 14$), employing a Think-Aloud Protocol, standardized questionnaires and semi-structured interviews. Quantitative results demonstrate very high system usability, achieving an average System Usability Score (SUS) of 88.93, alongside consistently low cognitive load metrics. Qualitative insights show that the translation of XAI outputs into executive summaries and follow-up interview questions helps bridge the actionability gap. Crucially, the interface fostered “calibrated trust”, leading to users actively questioning and verifying algorithmic outputs and thereby preventing automation bias. Finally, the thesis presents design guidelines for future XAI recruitment systems, highlighting the importance of data traceability and the integration of application documents into the candidate evaluation process.

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Introduction

1.1 Background: The rise of AI in Recruitment

Human Resource Management (HRM) is undergoing a major transformation driven by Artificial Intelligence (AI). Especially larger organizations receive an overwhelming volume of job applications, which makes manual applicant screening a labor-intensive bottleneck. To address this issue, companies are increasingly deploying AI-driven tools to make the application process more efficient, speeding up the hiring process and saving time [Abi25]. Examples for such tools include applicant tracking systems (ATS) [Abi25] and large language models (LLMs) [GZM24] to automate tasks like resume parsing, ranking and candidate assessment. Recent studies indicate that by 2025, almost 99 % of Fortune 500 companies had integrated some form of AI into their hiring processes to enhance efficiency and scalability [Abi25]. On a broader scale, the overall adoption of AI applications in talent management is projected to experience a steady upward trajectory, reaching approximately 70 % by the year 2030 [AGB⁺24].

However, while these technologies have seen rapid technological advancements and developments, their internal mechanisms remain invisible to their users [Fab25, TZY⁺25]. Most high-performing models operate as “black boxes” [Fab25], which means that the model’s internal input-output mapping remains opaque [TZY⁺25].

1.2 Problem Statement: The “Black Box” and the Need for Accountability

Opaque recruitment pipelines not only erode user trust in the final recommendations [TZY⁺25], but they also make it nearly impossible to audit the system for ethical bias, legal compliance, and predictive correctness. Without understandable outputs, it is increasingly difficult for applicants to comprehend the rationale behind their rejections.

From a recruiter’s perspective, an opaque system hinders employers from examining their application process for bias and regulators find it hard to confirm that decisions comply with anti-discrimination laws [Fab25]. By contrast, an explainable system leads to an improved recruitment efficiency. Moreover, a transparent application can help developers identify and optimize issues in algorithmic design [TZY⁺25].

Especially more complex models or deep learning algorithms offer high accuracy but lack traceability and accountability [OEUGOO⁺24]. This makes interpreting a model’s decisions difficult to impossible, reducing trust in the system [Fab25, TZY⁺25].

It can be observed that while modern complex models can accurately extract intricate patterns from unstructured text like resumes, their lack of transparency obscures how different features contribute to recommendations. This is a direct consequence of these models’ inner structure and particularly applies to models based on deep learning and attention architectures. The multilayer abstraction and nonlinear transformations of deep neural networks render their internal workings opaque, making it difficult to trace feature importance. In contrast, simpler models, such as decision trees, provide transparency through explicit rules. However, it has been observed that approaches using gradient-boosted decision trees can result in an approximately 10-15 % lower accuracy than deep models due to their limited ability to capture the semantic patterns found in complex resumes. Furthermore, existing legacy systems often rely on the automatic weighting of keywords extracted from job descriptions. These applications provide an opaque weight-assignment process, hiding each feature’s importance and further obscuring the composition of an applicant’s overall score [TZY⁺25].

Ultimately, the inability to trace how complex models evaluate candidate features leaves Human Resources (HR) professionals reliant on opaque scores [RIA25, TZY⁺25], severely limiting their ability to ensure fair and legally compliant hiring practices [Abi25, OEUGOO⁺24, Fab25].

1.3 The Ethical & Regulatory Constraint: Bias & Fairness

The opacity of AI systems poses critical ethical and practical challenges. Without transparency, it is difficult to detect algorithmic bias, ensuring that hiring decisions are fair regarding gender, race and age [Abi25]. Furthermore, European regulations such as the General Data Protection Regulation (GDPR) and the EU AI Act increasingly mandate explainable automated decisions, which can be subject to human oversight [Fab25].

Biased decision-making in hiring processes can have multiple origins, including the datasets used for training and human made design choices during system and algorithmic design. Also, AI systems may replicate biased patterns found in historical hiring decisions as well as human hiring decisions [Abi25]. A well-known example for this is an AI recruitment system that has been used by Amazon, which persistently promoted male over female applicants [Abi25]. In this instance, because the training data was overwhelmingly skewed

toward past male hires, the algorithm learned to actively downgrade CVs containing female-associated identifiers, such as participation in a “women’s chess club” [RIA25].

Ideally, a hiring system should be designed to be fair. However, fairness is difficult to define in reality as there exist different definitions and understandings of fair behavior [Abi25]. While it is mentioned in existing literature that this creates the need to include fairness as part of the recruitment process [Abi25], the problems described impede implementing a widely accepted fair system. To counter these issues, a system can be designed to be transparent, making it possible for the systems’ users to assess this fairness. To implement this transparency, the field of Explainable AI (XAI) has emerged. XAI techniques, such as SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations), aim to make algorithmic decisions transparent by highlighting which features contributed most to a candidate’s score [Fab25]. This transparency supports the alignment of AI recommendations with fair hiring processes [RIA25].

Users evaluate AI systems not purely on their predictive accuracy, but largely on whether the generated outcomes are transparent and equitable. While introducing explainability features can increase user trust, transparency alone is often insufficient to prevent users from blindly accepting AI outputs. Additionally, XAI systems might suggest fairness without actually improving design quality or mitigating discrimination [Fab25]. Additionally, XAI systems can indicate fairness without actually improving design quality or reducing discrimination [Fab25]. This highlights the distinction between transparent and fair systems: While XAI systems provide transparency to reveal potential inequity, they do not guarantee fairness themselves.

1.4 The “Actionability Gap” in XAI Applications

While recent literature extensively demonstrates the successful application of XAI techniques to reduce bias and provide transparency in HR systems [SKS23, HCR⁺22, OEUGOO⁺24, Fab25, RIA25, PMG25, ZPTY25, Abi25], these approaches primarily focus on identifying bias rather than providing concrete support for recruiters.

However, Fabeyo describes how current XAI methods can foster an “illusion of fairness” or “false fairness”. While tools like SHAP are mathematically precise, they are also semantically ambiguous. Because interpreting their outputs requires AI expert knowledge, these methods fail to empower end-users or improve actual decision quality [Fab25]. This demonstrates that showing the result of explainable techniques is not sufficient to create a fair system and shows the need for actionable applications. Moreover, counterfactuals are highlighted as a way to offer “actionable insights” [Fab25].

In a study conducted by Pinto et al., a vital socio-technical requirement for HR systems called “bargaining” is introduced. It describes that employees and candidates want to express, negotiate and contextualize AI outcomes to balance power asymmetries [PMG25].

Additionally, Tang et al. emphasize the need for human-in-the-loop pipelines and feedback-driven adaption. The review points out that systems need to move beyond static

explanations by allowing users to interact with and refine the explanations [TZY⁺25].

Finally, Naveed et al. describe how the “usability” of an explanation is measured by its usefulness and actionability. This shows that an explanation must directly enhance the user’s task performance and help them make better decisions [NSRK24].

While existing XAI solutions successfully address the questions of why a candidate was selected, they often stop short of supporting the subsequent human decision-making process. Current systems provide static explanations, but they rarely guide the recruiter on how to verify or explore AI-identified strengths, weaknesses or open questions during the actual interview [HCR⁺22]. This leaves a gap between the automated analysis and the human verification stage, forcing recruiters to manually bridge the divide between an abstract AI score and a concrete interview strategy.

1.5 Proposed Solution: Hybrid Framework

To address the “actionability gap” identified in currently existing XAI applications for HR, this thesis proposes and evaluates a novel hybrid framework. The system’s key objective is to provide clear insights for recruiters to allow active evaluation and questioning of the system’s recommendations. To fulfill this objective, a combination of traditional Machine Learning algorithms and LLMs for generating narrative summaries and follow-up interview questions is used.

The proposed pipeline integrates several modules to process and score applicants transparently. First, an LLM is utilized for resume parsing. As it has been shown that LLMs excel at processing unstructured raw text [GZM24], LLMs are used for extracting a candidate’s information into structured, machine-readable JSON format. This structured data is then passed to a Random Forest classifier for candidate scoring. Random Forests have previously been validated for XAI in the HR domain [CGC⁺25] as they allow for a highly reliable application of post-hoc explainability methods compared to deep neural networks or standalone LLMs. Additionally, the system incorporates dynamic weighting of features through an interactive interface, allowing the scoring mechanism to adapt during post-processing based on specific recruiter preferences and priority criteria, thereby satisfying the need for interactive, actionable XAI systems [Fab25].

To ensure accountability and regulatory compliance, the framework applies TreeSHAP to the Random Forest outputs. This provides the recruiter with a mathematically precise [Fab25] breakdown of which specific features influenced a candidate’s score. However, to further expand the system’s transparency and actively facilitate the socio-technical requirement of “bargaining” [PMG25], the application employs a subsequent LLM prompt to display the identified results on a comprehensive recruiter dashboard by translating the raw XAI outputs into a textual summary, visual feature-impact diagrams and targeted, actionable interview questions. By providing recruiters with specific, context-driven topics to clarify during an interview, the system directly supports task performance [NSRK24] and the subsequent phases of the hiring process.

Finally, to determine the practical viability of this hybrid approach, the proposed system will undergo a usability evaluation. This assessment will be conducted with a test group of university students to gather qualitative and quantitative insights.

1.6 Research Objectives and Contributions

This thesis aims to provide a novel, hybrid framework for supporting recruiters in the recruitment process. By implementing and evaluating a system that transforms static explainability metrics into human-centric insights, this system intends to provide tools for recruiters to actively verify and question the system’s outputs.

To achieve this goal, the following research objectives have been set:

1. **Prototype Development:** Design and implement a framework that combines LLMs for unstructured text parsing, a Random Forest classifier for candidate scoring and TreeSHAP for explainability. The implementation is split into an interactive dashboard, which allows dynamically weighting different features and the backend prompting the LLM to generative a narrative summary as well as follow-up interview questions and handle the logical processing and candidate scoring.
2. **System Evaluation:** Evaluate the usability and usefulness of the proposed framework and measure user trust in the system by conducting a user study with university students. This evaluation focuses on testing whether the generated insights assist recruiters in a Human-in-the-Loop (HITL) decision-making process.
3. **Derive Guidelines:** Derive and formulate a set of guidelines for the design and deployment of XAI tools from the literature review, implementation and user study.

By fulfilling these objectives, this thesis makes several contributions to the deployment of XAI in the HR field:

1. **Theoretical Contribution:** A detailed exploration of the “actionability gap” identified in HR XAI literature, demonstrating the necessity to move beyond passive transparency toward systems that actively support human-in-the-loop decision-making.
2. **Artifact Contribution:** The development of a functional, hybrid prototype, that successfully combines traditional XAI with generative AI to not only rank candidates but to actively assist in the subsequent interview stages.
3. **Empirical Contribution:** Collected insights from the user study, which examine the usability and usefulness of the proposed system.

4. Practical Contribution: A formalized set of guidelines to assist researchers, developers and HR professionals in designing transparent, user-centric AI recruitment architectures.

Related Work

2.1 Existing Domains of XAI Applications

Ullah et al. demonstrate how XAI techniques can help evaluate and broaden existing knowledge as well as expanding knowledge boundaries and create new ideas. They have also shown how XAI successfully achieves six key goals across industries: transparency, trust, understanding, accountability, ethics and innovation. The study highlights successful applications in critical domains like autonomous medical agents, autonomous vehicles, threat detection and finance [UKDFS24]. These applications demonstrate that XAI is utilized for high-stakes decision-making. Furthermore, Brasse et al. mention that XAI can be used in information systems to evaluate systems for flaws, improve model accuracy, reason for accountability and reveal new relationships from underlying data [BBF⁺23]. Additionally, Patil describes a reduction in bias, an assessment of model reliability and improved decision-making via AI system transparency as consequences of XAI in industry applications. Moreover, XAI helps organizations comply to regulatory standards and strengthen consumer and stakeholder confidence [Pat24].

2.2 XAI and Automated Decision-making in Human Resources

Within the HR industry, XAI applications are increasingly deployed to effectively handle overwhelming numbers of applications [Abi25] while balancing mathematical precision [Fab25] with human interpretability [TZY⁺25]. For instance, Chaudary et al. combined Machine Learning models, including Random Forests and a decision tree classifier, with SHAP to derive feature values to predict employee turnover. As churn prediction depends on multiple factors, assessing and forecasting this turnover is critical. Thus, proactive steps to reduce turnover risk can be taken based on the interpretable results derived. From the models tested, Random Forest and Logistic Regression performed best, achieving an

accuracy of 87.29 % and 87.96 % respectively [CGC⁺25], validating both Random Forests and Logistic Regression for HR predictions. While Logistic Regression was found to be marginally more accurate (by less than 0.7 %), Random Forest stands out because of its perfect architectural synergy with TreeSHAP. Random Forest’s ensemble of decision trees allows SHAP to computationally efficiently calculate exact feature attributions [CGC⁺25], which is critical for generating the actionable interview questions required by this system. More specific to candidate selection, Bhattacharya proposed “AaJeeViKa”, an AI-based recruitment scheme employing Machine Learning for candidate scoring and successfully applies SHAP explainers to validate and interpret those predictions [BZR⁺22]. This work further establishes the application of post-hoc SHAP analysis to candidate scoring methods as a valid and effective approach to ensure transparency in HR systems. Furthermore, post-hoc explanation tools like LIME and SHAP are highly effective for providing thorough evaluations of feature contributions without sacrificing overall predictive accuracy [UKDFS24]. This explicitly validates the selection of SHAP for this proposed architecture, as it ensures clear insights into complex algorithmic choices within the high-stakes domain of recruitment.

Recent studies demonstrate that XAI can be utilized not merely for post-hoc application, but as an active component in the pipeline. Sogancioglu et al. introduced the “ProxyMute” algorithm, which trains a secondary classifier to predict sensitive attributes and applies TreeSHAP to identify feature importance. Features with high SHAP values are flagged as bias proxies and are computationally excluded during inference. While this approach has been used to actively input data to improve fairness without retraining [SKS23], this thesis extends the concept of actionability by using SHAP output to generate dynamic, Human-in-the-Loop interview questions.

Additionally, Hofeditz et al. provide empirical evidence through an online experiment including 194 evaluated responses, revealing that AI-driven recommendations effectively lowered historical biases against older and female applicants, though they paradoxically led to a decreased selection rate for foreign-race candidates. However, the study also highlighted that recommendations alone can sometimes trigger algorithmic aversion, a behavioral tendency to dismiss algorithmic predictions, against foreign-race candidates if the AI’s preference seems overly obvious. Introducing XAI-induced transparency positively moderated these effects, helping recruiters reflect on their biases and improving the acceptance of diverse candidates [HCR⁺22]. By using the SHAP values to identify interview questions, the system introduced in this thesis expands rating a candidate by a more actionable interaction. Furthermore, the system actively forces the recruiter to reflect on discrepancies by allowing recruiters to dynamically weight features and comparing their weighted score to the system’s baseline score, therefore impelling recruiters to actively question and verify the system’s outputs.

Evans-Uzosike et al. show that XAI enables HR teams to identify and mitigate algorithmic bias. They highlight that techniques like SHAP help detect hidden patterns of discrimination, which allows data scientists to implement corrective weighting or feature engineering. Furthermore, it is described that although algorithmic bias cannot

be fully neutralized, it can be minimized by validation and design. XAI systems allow for the diagnosis, correction and communication of errors, encouraging accountability and ensuring that decisions reflect organizational values [OEUGOO⁺24].

In a systematic review of transparency methods in hiring algorithms conducted by Fabeyo, methods such as SHAP, LIME, decision trees, and counterfactuals are identified and assessed regarding interpretability, fidelity and fairness. Transparent systems can improve the understanding and acceptance of AI systems, but can be insufficient to diminish dependence on incorrect predictions. Furthermore, studies indicate that individuals often place excessive reliance on AI-generated advice, a behavior that persists even when the system’s reasoning is fully transparent. However, trust and decision quality are found to be positively influenced by recommendations, allowing to audit fairness, improve understanding and reduce biases in employment decision-making [Fab25]. To bridge the gap between transparency and real-world deployment, additional features to support the independence of those predictions are necessary. This is facilitated by generating both follow-up interview questions and textual summaries as well as the candidate ranking. Additionally, the illusion of “false fairness” can be caused by oversimplified or misleading explanations. Therefore, the output of XAI applications must be assessed for human comprehension and its effects on behavior and trust [Fab25]. Consequently, human understanding metrics need to be evaluated in the user study. Moreover, Fabeyo demonstrates that while models, which are explainable by design, provide intuitive frameworks for reconstructing model choices, they do not fully reflect complex data trends. Also, a downside of tools like SHAP is identified: While explanations work well for candidates following the majority of the data, they can lead to unreliable predictions if candidates diverge from the standard [Fab25]. In order to properly test the system introduced, it needs to be assessed using applicants, which differ from an average job applicant. In addition to these requirements, the handling of missing values needs to be addressed in the implementation.

According to Riaz, AI-driven recruitment algorithms are susceptible to several distinct forms of ethical risk. Specifically, these systems can inherit historical data bias from past human decisions, suffer from feature selection bias by favoring established historical profiles or produce skewed outputs due to imbalanced and non-diverse training datasets [RIA25]. Thus, the dataset choice, including its composition and cultural diversity, and the system’s ability to handle applicants who do not fit past training data, must be considered.

Zhang et al. review the application of XAI across different job application stages. They note that generative models for skill-related interview questions have already been developed, which employed a retrieval system that provides personalized question recommendations for interviewers. Additionally, a graph-enhanced retrieval system was used, which returned interview questions based on a specific set of skills. Research shows that informational fairness can be enhanced using AI interviewers, which provide “high-informativeness feedback”. In talent recruitment processes, bias can lead to inequalities in minority groups and gender gaps. Interviewers perceive a higher fit and provide

better interview assessments if candidates' genders correspond to gender stereotypes. Furthermore, LLMs are susceptible to perpetuating historical biases present in their training datasets [ZPTY25]. This further justifies the hybrid architecture chosen for this thesis, which restricts the LLM to parsing and generation tasks, while reserving the actual candidate scoring for a Random Forest model, which can be audited. Moreover, existing XAI methods are critiqued as they “lack real-time response capabilities to bias evolution in dynamic recruitment scenarios” [ZPTY25]. By dynamically generating interview questions from identified SHAP values, the introduced system allows recruiters to respond to potential algorithmic biases in real-time.

A comprehensive review by Abidizadegan demonstrates that general-purpose LLMs consistently fail statistical parity tests when scoring candidates. Studies show they often exhibit significant racial and gender biases based purely on candidate names [Abi25]. This further reinforces the architectural decision of this work to restrict the LLM to parsing and question generation, while delegating candidate scoring to a transparent Random Forest model. Furthermore, Human-in-the-Loop frameworks are found to be essential for equitable AI recruitment [Abi25]. The system proposed operationalizes this requirement by translating SHAP scores into targeted interview questions and allowing recruiters to dynamically weight features. This ensures the AI acts as an advisory tool, which assists human judgment, rather than replaces it. Moreover, it is highlighted that current research lacks sustainable, real-world strategies to mitigate algorithmic bias [Abi25]. By shifting from passive bias detection to active bias review, which is enabled by generating questions to verify AI scores and answer identified gaps, this thesis presents a practical, real-world framework. Additionally, it is stated that the GDPR, a European data protection regulation, grants candidates the right to obtain an explanation regarding automated decisions. Similarly, the EU AI Act categorizes hiring algorithms as “high-risk”, mandating that these tools have reliable data, regular fairness testing, clear explanations for decisions and human oversight [Abi25]. The introduced pipeline aligns with these regulatory claims: it fulfills the EU AI Act’s requirement for transparency and human oversight and the GDPR’s mandate for explainability. Lastly, the review notes that “feature selection bias” occurs when seemingly neutral data points act as proxies for sensitive traits like race or socioeconomic background. It is emphasized that XAI methods allow recruiters to understand how systems make decisions so that they can easily identify and correct hidden biases in resume screening [Abi25]. Consequently, the system’s SHAP values allow recruiters to visualize if the Random Forest is inappropriately weighting proxies, giving them the opportunity to uncover and handle unfair rankings.

Brasse et al. conducted a literature review, which highlights a major research gap: the current focus on novel algorithms rather than the creation of “human-consumable explanations” that address specific user needs [BBF⁺23]. While raw SHAP values fail to meet this requirement, translated natural language interview questions facilitate this requirement. Furthermore, the literature highlights a need for user-driven XAI tools where individuals can manually investigate how the algorithm operates and adjust the level of transparency to match their personal context [BBF⁺23]. By including the

possibility to dynamically adjust the weighting of different skills, the introduced system addresses these demands: it provides customized explanations per user and skills needed and is interactive by design. Moreover, Brasse et al. call for a more diverse mix of “human-grounded” evaluations. They have identified that 71 % of studies ignore human factors [BBF⁺23], which validates the assessment using a user study conducted with university students. Finally, it is noted that explainable information systems enable stakeholders to evaluate systems for flaws, improve model accuracy, justify decisions for accountability and learn new causal relationships from underlying data [BBF⁺23].

Despite the advancements in utilizing XAI for HR applications, current tools, such as SHAP and LIME, frequently generate complex, mathematical feature attributions that are notoriously difficult for non-technical HR employees to interpret [DRWN25, SVG⁺25]. This creates a phenomenon where explanations are built by technical experts tailored for their own understanding, rather than for the end-user [DRWN25]. While recent literature has introduced “XAI narratives” to translate these values into text, relying solely on the passive textual explanations can foster an “illusion of fairness”, where users over-rely on the system’s recommendations without understanding or challenging the underlying biases [Fab25]. Consequently, there is a critical gap in shifting XAI from a passive reading experience to actionable tool that actively empowers the recruiter’s independent judgment.

Furthermore, existing XAI methods in the recruitment domain are heavily critiqued for lacking real-time response capabilities to bias evolution in dynamic hiring scenarios [ZPTY25]. The literature explicitly calls for the development of interactive XAI solutions that allow users to adapt and explore an algorithm’s behavior dynamically [BBF⁺23]. Few existing tools enable HR professionals to interactively adjust feature weights, limiting their ability to audit systems for feature selection biases or proxy variables [Abi25, SKS23]. Finally, a persistent methodological flaw in current XAI literature is the over-reliance on functionally grounded evaluations (purely technical testing) at the expense of human-centric testing. Research indicates that 71 % of studies developing novel XAI approaches completely neglect evaluation with human users [BBF⁺23]. Within the specific context of the labor market, there remains a gap regarding the assessment of user satisfaction and the practical comprehension of XAI tools by non-technical stakeholders [PMG25, Fab25].

2.3 LLMs in Resume Parsing and Text Generation

2.3.1 LLMs in Resume Parsing

Historically, resume parsing, which is the process of converting unstructured resume text into standardized, machine readable formats, relied on rule-based matching, regular expressions and traditional machine learning algorithms like Hidden Markov Models and Support Vector Machines [YGZ05, ZPTY25]. These computationally lightweight models struggled to reliably parse real-world applications due to their highly inconsistent formatting and varied phrasing [ZYC⁺25]. More advanced methods, like pre-trained Language Models (PLMs), such as BERT and RoBERTa, were adopted to classify

individual resume sentences [GM22]. However, strict input length limitations restrict their applicability, making them unsuitable for processing entire resumes [GZM24].

The development of modern LLMs has significantly advanced automated resume screening. Previous machine learning algorithms require extensive manual feature engineering and encounter degraded performance when handling complex, large-scale datasets. In contrast, LLMs possess the ability to process completely unstructured textual content effectively [MS24]. These models leverage broad pre-training to capture nuanced language, infer missing context and manage longer texts. Consequently, they enable the simultaneous evaluation of an entire resume [GZM24, MS24].

Recent literature investigates the use of LLMs as autonomous agents to extract and structure resume data. Gan et al. proposed a fully automated LLM-agent framework that utilizes an instruction-tuned LLaMA2 model. This model classifies resume sentences and filter out sensitive personal data for privacy preservation. Subsequently, the system employs GPT models acting as “HR Agents” to parse the remaining text into standardized JSON formats, generating brief summaries and grades. This LLM-based approach proved to be up to 11 times faster than manual screening while closely imitating human evaluative patterns [GZM24]. Similarly, systems like “Resspar” leverage generative AI application programming interfaces (APIs) alongside traditional natural language processing to robustly extract structured information directly from raw resumes. This includes job titles, work experience and technical abilities [SGPC25].

The deployment of LLMs for resume parsing tasks introduces challenges, such as the heterogeneity of multi-column resume layouts, high inference latency and the risk of generative hallucinations [ZYC⁺25]. Recent industrial frameworks integrate layout-aware pre-processing with optimized LLM prompting to increase performance. Zhu et al. introduced a unified pipeline that uses a fine-tuned object detection model (YOLOv10) to identify complex layout segments and serialize them into a semantically coherent, indexed sequence. To optimize the LLM’s efficiency and eliminate content drift, they implemented an “index-based pointer mechanism”. Instead of prompting the LLM to generate full verbatim descriptions, which is token-intensive and prone to hallucinations, the model is instructed to output only the line index numbers referencing the original text, outputting the final structured data as a JSON object [ZYC⁺25].

While the extraction of capabilities of LLMs are state-of-the-art, deploying them in operational HR environments introduces severe challenges regarding data privacy and computational resource constraints [Sat25]. As resumes contain sensitive personal data protected by regulations such as the GDPR, transmitting these documents to proprietary cloud APIs presents significant compliance risks [Sat25, NLL⁺26]. Nawalny et al. propose a privacy-preserving framework that converts PDFs to Markdown and applies deterministic anonymization protocols to prevent sensitive identifiers from leaving the processing environment [NLL⁺26]. Furthermore, research shows that massive proprietary models are not strictly necessary to achieve precise extraction. Satıcı highlights that Small Language Models (SLMs), such as Llama-3.2-1B, can be fine-tuned to extract context-dependent fields, which traditional system consistently fail to parse, while efficiently running on

“commodity hardware” [Sat25]. To ensure these models generate the strict, machine readable JSON formats required for machine learning tasks, researchers emphasize the necessity of deterministic decoding, such as a temperature of 0.0, combined with rigorous schema validation [NLL⁺26]. Consequently, the LLM chosen for this thesis will need to be utilized using deterministic decoding using a temperature of 0.0.

These recent advancements establish an academic and practical foundation for utilizing LLMs in resume parsing. The ability of LLMs to handle diverse document layouts and convert heterogeneous, unstructured text into standardized JSON formats facilitates this extraction process. Therefore, this literature justifies the architectural choice of this thesis: employing an LLM as the primary parsing engine enables the transformation of unstructured resumes into structured skill sets and experiences. This structured extraction is a requirement for feeding features into the downstream interpretable models to evaluate candidates and generate narrative summaries and dynamic interview questions.

However, while LLMs demonstrate state-of-the-art proficiency in parsing unstructured textual content [GZM24, MS24], utilizing them to directly score or rank candidates introduces severe ethical risks. Recent systematic reviews demonstrate that general-purpose LLMs consistently fail statistical parity tests and frequently exhibit significant racial and gender biases based purely on candidate demographics or names [Abi25]. Furthermore, employing an LLM as an autonomous explainer without direct access to an underlying model’s computational process is unreliable and highly susceptible to hallucinations [ZYC⁺25]. Current research lacks frameworks that successfully deploy the semantic power of LLMs while strictly preventing them from executing biases, non-deterministic candidate scoring.

2.3.2 LLMs in Text Generation

Previously, natural language generation in specialized domains relied heavily on hand-engineered templates or early sequence-to-sequence neural architectures to synthesize information and generate questions [LZC20, QZS⁺23]. While these approaches successfully produced structured outputs, they lacked the semantic flexibility required for dynamic human-computer interaction. Similarly to resume parsing, the establishment of LLMs revolutionized this space, enabling “XAI Narratives” – a novel paradigm, which is marked by utilizing LLMs as post-hoc translators to convert complex, technical XAI outputs into coherent, human-readable narratives [SVG⁺25]. Traditional XAI techniques, such as SHAP, output complex, mathematical feature attributions that are difficult for non-technical stakeholders, such as HR employees, to interpret. This creates a gap between technical transparency and practical human interpretability [SVG⁺25, DRWN25, ZPV24].

Recent literature shows that LLMs excel at bridging this interpretability gap. For instance, frameworks like “SHAPstories” employ LLMs to translate tabular SHAP values into natural language narratives. Studies show that over 90 % of users find these narrative explanations convincing and easier to interpret than standard SHAP waterfall plots [MHD⁺25]. Similarly, the “ContextualSHAP” package integrates OpenAI’s models to

extend raw SHAP visualizations using contextual generative text, proving that combining visual data with tailored textual summaries significantly increases a user’s perceived understandability and confidence in the system’s reasoning [DRWN25]. Furthermore, research demonstrates that smaller, highly optimized LLMs can achieve exceptional qualitative performance in generating complete, fluent, and context-aware XAI narratives when guided by structured prompts, such as zero-shot or few-shot examples [MC26, ZPV24]. These findings support the inclusion of a textual candidate summary of XAI values in this thesis. Converting algorithmic scoring into a natural language narrative has been shown to improve recruiter trust and comprehension [MHD⁺25, SVG⁺25, ZPV24].

Beyond summarizing data, LLMs demonstrate reasoning capabilities that allow them to dynamically generate targeted interview questions based on contextual inputs. While early systems like DuerQues relied on specific knowledge-sharing community data to build sequence-to-sequence question generators [QZS⁺23], modern LLMs can perform this task dynamically and in a zero-shot or few-shot setting. Recent studies evaluating LLMs for automated leadership competency assessments demonstrate that models can analyze candidate profiles and generate behaviorally relevant follow-up questions tailored to specific competencies [GARA26]. Furthermore, advanced prompt engineering techniques, such as Chain-of-Thought (CoT) and few-shot-prompting, have been proven to empower LLMs to reliably produce diverse academic assessments tailored to varying degrees of cognitive difficulty [SDCS24]. This validates the architectural choice of utilizing an LLM to generate follow-up interview questions tailored to the exact strengths or gaps identified in a candidate’s SHAP profile.

2.4 Summary of Research Gaps and Thesis Contributions

As identified throughout the preceding literature review, the current integration of AI into HR faces several research gaps. Chief among these are the reliance on passive XAI summaries that create an “illusion of fairness” [Fab25], the inherent biases and unreliability of utilizing LLMs as direct candidate scorers [Abi25], a lack of real-time interactive bias review mechanisms [ZPTY25] and a paucity of human-grounded evaluation methodologies [BBF⁺23].

This thesis differentiates itself from existing literature by proposing a novel, end-to-end hybrid architecture designed specifically to address these systemic gaps. The contributions of this work are defined through the following key differentiators:

- From Passive Summaries to Actionable Interview Questions: To overcome the “illusion of fairness” [Fab25], the proposed system translates raw SHAP values directly into targeted, dynamic interview questions base on the exact strengths or gaps identified in a candidate’s profile. This shifts the AI’s role from a direct decision-maker to a collaborative workflow assistant, supporting the recruiter in an active human-in-the-loop workflow [Abi25] by verifying the AI’s reasoning.

- **Transparent Hybrid Architecture:** Addressing the inherent biases of LLM-based scoring, this thesis implements a strict separation of concerns. The LLM is restricted exclusively to deterministic resume parsing and question generation, while the high-stakes task of candidate scoring is delegated to a mathematically auditable Random Forest model paired with TreeSHAP [CGC⁺25]. This hybrid “Glass-Box” pipeline ensures transparency while leveraging the advanced semantic capabilities of LLMs [OEUGOO⁺24].
- **Dynamic Interactivity:** Answering the academic call for interactive XAI [BBF⁺23], the system introduces dynamic feature weighting. Recruiters can manually adjust the importance of different skills and observe how each candidate’s ranking changes in real-time. This interactive capability acts as a manual counterfactual audit [ZPTY25], enabling an information badge for candidates with strongly deviating scores. These applicants may have been ranked high due to the recruiter’s dynamic scoring of skills, but have a significantly lower system score. This directly addresses the lack of real-time response capabilities in current systems.
- **Human-Grounded Evaluation:** Differentiating from studies that rely solely on technical metrics, this thesis validates the proposed system through a human-grounded evaluation [BBF⁺23, NSRK24]. By conducting a user study and gathering direct feedback from testers, the evaluation specifically measures how humans interact with the system and investigates user satisfaction with the usefulness of the system.

Methodology

3.1 System Architecture

To translate passive XAI outputs into practical support and insights, the proposed system is implemented using a hybrid architecture. The pipeline is designed to process applicant data, score the candidates and enable Human-in-the-Loop interaction by providing an interactive recruiter interface. The processing flow of the architecture consists of four stages: document ingestion, data extraction, candidate scoring and the generation of candidate summaries and interview questions.

3.1.1 Document Ingestion and Initial Weighting

The system initiates through a dual-upload strategy, requiring the recruiter to provide a job description and the candidate resumes. Before processing the candidates, the system analyzes the job description to establish a baseline for the evaluation criteria. This is essential for initializing the dynamic feature weights. By extracting the core requirements from the job listing, such as years of experience or required education, the system automatically sets the corresponding feature weights. This ensures that the initial ranking criteria represents the requirements of the job description rather than arbitrary default values.

Initially, recruiters are required to upload a job description as well as candidate CVs to the system. The system first processes the job description to establish a baseline for the evaluation criteria. Specific requirements are extracted to automatically pre-adjust the corresponding feature weights in the dashboard. Consequently, the initial ranking of candidates reflects the requirements of the job listing rather than default values.

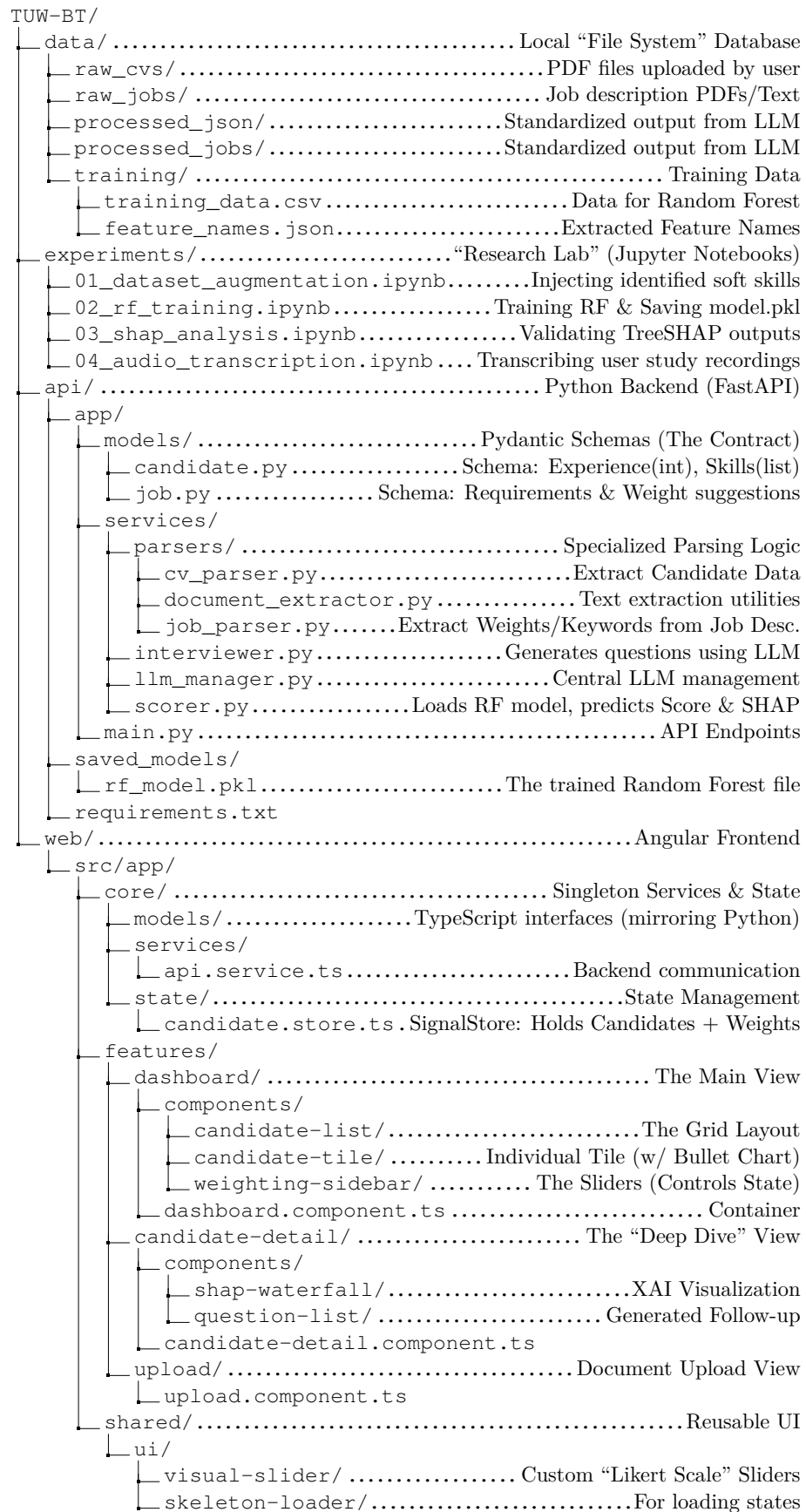


Figure 3.1: Logical repository structure separating data, experimentation, and application layers.

3.1.2 Deterministic Data Extraction

Once the baseline weights are established, the unstructured candidate resumes are processed using an LLM. To prevent generative hallucination and ensure strict validation, the model’s temperature is set to 0.0 to ensure deterministic decoding during the extraction phase. The LLM parses the resumes and converts the unstructured text into standardized, machine-readable JSON formats. Each candidate’s experiences and skills are mapped uniformly to the features required by the scoring model.

3.1.3 Dual-Scoring Mechanism and Inherent Ranking

The system’s core recruiter interaction relies on the inherent candidate ranking system, which is driven by a dual-scoring mechanism. Instead of presenting single, opaque recommendations, the architecture calculates two distinct metrics:

- **Objective Random Forest Score:** A baseline probability score representing the Random Forest classifier’s unweighted score. This score depicts the evaluation of a candidate based on the training data taken from the dataset.
- **Subjective User Score:** A dynamically calculated score that reflects the recruiter’s current weighting of specific features. The user score is initialized to the initial weighting described above.

Using the interactive dashboard, recruiters can adjust the importance of different features via Likert-scale sliders (ranking from “Ignore” to “Critical”). Instead of applying these subjective weights to raw, unnormalized feature values, the architecture integrates the recruiter’s preferences directly into the model’s XAI framework.

The system recalculates each candidate’s User Score by modifying the individual SHAP values. Specifically, the model’s baseline expected value is adjusted by the sum of the candidate’s SHAP feature attributions, each scales by a multiplier derived from the recruiter’s slider input:

$$UserScore = BaseValue + \sum_{i=1}^n (SHAP_Value_i \times UserMultiplier_i) \quad (3.1)$$

The candidates are subsequently ranked and ordered on the dashboard based on this dynamically calculated *UserScore*. This approach places the applicants who best match the recruiter’s specific, weighted criteria at the top of the interface, while ensuring the underlying score remains mathematically bounded by the model’s objective evaluation of feature importance.

3.1.4 Score Deviation Flag and XAI Generation

To prevent algorithmic aversion and mitigate the “illusion of fairness” often encountered in passive XAI systems, the architecture actively compares the two scores to generate score deviation badges. If a candidate achieves a high weighted score but the underlying Random Forest score falls below a predefined safety boundary, the system visually flags the candidate. This acts as additional information, visualizing that while the candidate matches a recruiter’s subjective criteria, the objective model ranks the candidate significantly lower.

Finally, the underlying Random Forest model is audited using TreeSHAP to calculate exact feature attributions. These raw SHAP values, alongside the identified score deviation flags and candidate features, are used to create a visual feature impact diagram and are subsequently passed to the LLM using a secondary prompt. This final module translates the mathematical attributions into a natural language executive summary and generates targeted, actionable interview questions. These questions provide the recruiter with specific topics to verify or discuss during the interview.

3.2 Architectural Choices and Justification

3.2.1 Random Forest Classifier

While deep learning models, such as CNNs and RNNs, achieve high predictive accuracy in recruitment tasks, their multilayer abstractions obscure feature importance, resulting in an opaque system and making them unsuitable for high-stakes decision-making [TZY⁺25]. To prioritize both predictive accuracy and structural transparency, this system delegates the core task of candidate scoring to a Random Forest classifier. As an ensemble learning algorithm, it aggregates the outputs of numerous individual decision trees, making it highly capable of handling complex, non-linear HR data while actively preventing the risk of overfitting [Don24, YAH⁺25].

Research indicates that Random Forest algorithms perform effectively in automated resume screening and candidate suitability classification, achieving accuracies of up to 96 % without relying on black box architectures [YAH⁺25]. Furthermore, comparative analyses in the HR domain demonstrate that Random Forests balance predictive performance and transparency. These models estimate complex data patterns accurately while utilizing an ensemble of simpler decision trees that facilitate post-hoc interpretability [CGC⁺25, UKDFS24]. Consequently, the baseline score is derived from established classification process.

3.2.2 TreeSHAP Integration

To make the system transparent by providing insights into the scoring mechanism, TreeSHAP is applied to the Random Forest classifier. SHAP, which is rooted in cooperative game theory, calculates the marginal contributions of each input variable and assigns

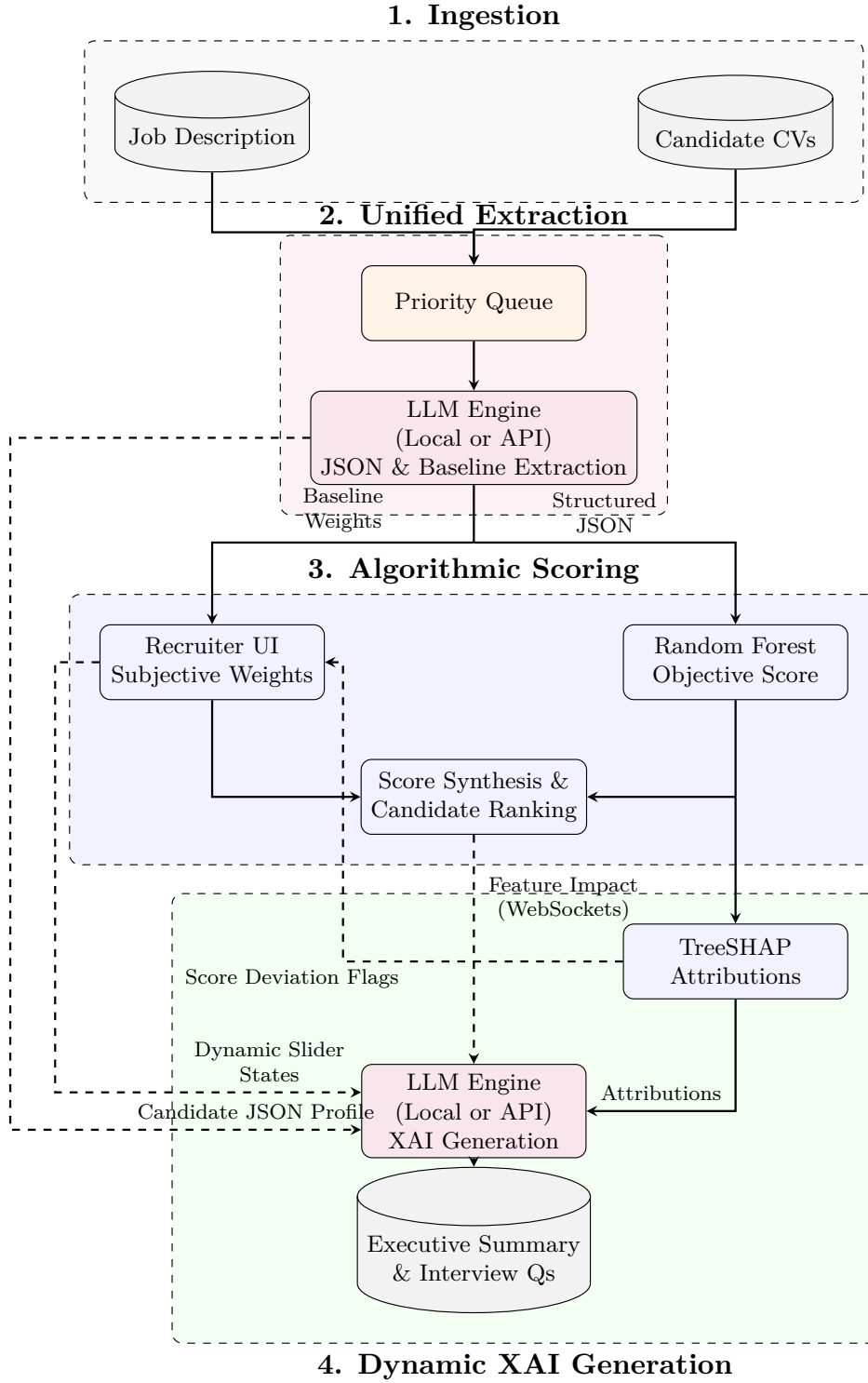


Figure 3.2: System architecture pipeline outlining the transformation of unstructured documents into ranked candidates and dynamic XAI narratives.

them an importance score. In this way, the impact of each input on the final score can be quantified [CGC⁺25, OEUGOO⁺24].

SHAP is paired with a Random Forest classifier due to their architectural compatibility. When applying SHAP to deep learning models or general-purpose algorithms, computationally expensive and potentially unstable approximations are required. Conversely, the use of a tree model allows the implementation of TreeSHAP, a specialized algorithm designed to leverage the internal structure of decision trees. Consequently, the calculation of exact SHAP values is performed in low-order polynomial time [DRWN25].

By guaranteeing exact and efficient feature attributions [SKS23], TreeSHAP forms the mathematically auditable bridge between the Random Forest’s score and the subsequently used, partially generated content. It ensures that the data passed to the LLM is grounded in statistical logic, establishing a mathematically precise foundation for the generated interview questions and textual summaries.

Building upon the distinction between automated fairness and system transparency established in Section 1.3 [Fab25], this architecture explicitly does not enforce algorithmic fairness or automated debiasing. Instead, the pairing of a Random Forest classifier with TreeSHAP serves as the technical realization of structural transparency [CGC⁺25, YAH⁺25]. By exposing the exact mathematical feature attributions, the system empowers the recruiter to evaluate and adjust the baseline logic via the dynamic interface [Abi25, HCR⁺22].

3.2.3 Flexible LLM Deployment for Structured Data Extraction

While massive proprietary LLMs, such as GPT-4o, excel at general text generation, and guaranteeing near-perfect schema adherence for data extraction, relying exclusively on them for automated resume screening introduces severe privacy concerns and regulatory compliance risks [NLL⁺26, Sat25]. To counter these risks while maintaining operational flexibility, this architecture provides a user-selectable deployment strategy, allowing recruiters to choose between using a cloud-based API and local, on-device deployment of open-weight SLMs, ranging from 1 to 8 billion parameters [Ale25, Sat25]. Processing recruitment documents requires the model to perform two different tasks simultaneously: it must convert unstructured text into machine-readable JSON formats without hallucinating data while also analyzing deep semantic reasoning to understand implicit soft skills and priority indicators within job descriptions [BDB⁺25, Sat25]. By offering both options, the system balances the reliability and extraction capabilities of cloud-based models and GDPR compliance enabled by local models like Llama-3.1-8B [GARA26, NLL⁺26]. To satisfy the privacy concerns raised by the GDPR, the proposed system employs an institutional, self-hosted infrastructure to ensure personally identifiable information (PII) remains strictly on-premise.

3.2.4 Flexible LLMs for Dynamic XAI Generation

Beyond data extraction, the framework must synthesize natural language narratives and actionable interview questions from mathematical XAI outputs [SVG⁺25]. This generative phase demands significantly higher semantic reasoning capabilities to produce context-aware text [GARA26, SDCS24]. Recent comparative analyses in natural language generation demonstrate that while state-of-the-art proprietary models offer maximum semantic expressiveness, open-source models in the 7 to 8 billion parameter range offer a competitive balance between robust, context-aware text generation and the computational efficiency required for local deployment [BDB⁺25, SDCS24]. To accommodate varying constraints, the framework allows the user to dynamically select the generation engine [GARA26]. Recruiters can route the XAI translation prompts to an API for superior generative capabilities or utilize a local model to ensure that sensitive data remains on-premise [GARA26, SVG⁺25]. Therefore, depending on the user’s preference regarding the trade-off between computational capability and data sovereignty, both model architectures can effectively function as the system’s “XAI narrative” generator [GARA26, NLL⁺26].

3.2.5 Context-Aware Prompting and Adaptive Generation

To ensure the generated interview questions are genuinely actionable, they cannot rely solely on the static mathematical outputs of the Random Forest model. A significant limitation of traditional XAI is the generation of static explanations that do not incorporate post-deployment user feedback, frequently resulting in persistent user distrust when the AI’s logic misaligns with the recruiter’s specific priorities [TZY⁺25]. To address this, the LLM employs a dynamic context prompting mechanism.

Rather than merely translating raw SHAP values into text, the generation prompt dynamically fuses three distinct data streams:

1. The candidate’s structured JSON profile (e.g. education, experience, skills).
2. The baseline TreeSHAP feature attributions.
3. The recruiter’s current subjective slider states (the dynamic feature weights).

This design choice aligns with the “ContextualSHAP” paradigm, which demonstrates that combining quantitative SHAP outputs with user-defined contextual parameters significantly improves perceived understandability and relevance [DRWN25]. Furthermore, as established by Silvestri et al., an effective XAI narrative prompt must act as an explicit conditioning variable that embeds user intent and situational cues [SVG⁺25]. By ingesting the recruiter’s subjective weights, the LLM is forced to contextually adapt its output, guaranteeing that the generated interview questions focus heavily on the specific traits the recruiter deems relevant.

3.2.6 Asynchronous Pipeline and Architectural Flow

Recent literature highlights that deploying LLMs within synchronous architectures blocks the main application flow, limiting system concurrency and causing user interfaces to freeze [GLZ24]. To balance high-level semantic performance with the required operational speeds, this architecture implements an event-driven, asynchronous processing pipeline [YZZ⁺25, GLZ24]. By decoupling the LLM’s inference from the frontend, the system enables non-blocking, independent operation that maximizes resource utilization and reduces perceived end-to-end latency [GLZ24].

The system deploys this concept through a background processing queue. As soon as a user uploads a batch of resumes, the files are instantly dispatched into this queue rather than being processed on the main thread. To maintain system responsiveness during the LLM’s high-latency extraction phase, the frontend instantly renders a “skeleton tile layout”. As the queue completes the parsing of each individual resume, real-time events push the extracted data to the frontend to populate the skeleton tiles. This continuous visual feedback is crucial for ensuring an intuitive user experience during human-AI interaction [NSRK24].

The asynchronous architecture is subsequently used for generating the candidate summaries and interview questions. To reduce the generation latency of LLMs [YZZ⁺25, Sat25], the system pre-generates this content in the background. While this pre-generation might get discarded if the feature weights get updated in the future, the overall fluidity and a responsive user interface are ensured [GLZ24, GKL⁺24].

This uninterrupted workflow is allows for maintaining the momentum required for an effective human-AI bargaining process [PMG25]. During standard operation, the background queue processes candidates based on the user’s initial upload order. However, user interactions dynamically manipulate this queue. If a specific candidate’s detailed profile is expanded, their question and summary generation are prioritized at the front of the queue. Similarly, adjusting the importance sliders triggers a regeneration of existing summaries. This fluid architectural flow satisfies the socio-technical requirement of bargaining by seamlessly adapting the AI’s inquiry strategy to the human’s evolving evaluation criteria in real-time, effectively reducing the feedback loop between human preference and algorithmic explanation [TZY⁺25].

3.2.7 Handling Missing Data in CV Extraction

A critical challenge in processing unstructured resumes is the frequent omission of standard data fields by applicants. While traditional machine learning pipelines frequently employ Missing Value Imputation (MVI) techniques, such as a Mean/Mode substitution or k-Nearest-Neighbors (kNN), to mathematically infer missing values [JSG24], applying these techniques to sensitive, context-dependent HR data introduces severe risks of data fabrication.

To maintain the integrity of candidate profiles, this system distinguishes between numerical

heuristics and categorical strings. For numerical attributes, such as *years_of_experience* or *projects_count*, missing values are defaulted to zero under the assumption that the omission of these fields signifies a lack of prior experience.

In contrast, string-based fields, such as *name* and *education* cannot be statistically imputed without generating hallucinations. Therefore, the proposed system adopts a “Structural Omission” strategy during the extraction phase. Following deterministic parsing protocols, the SLM is strictly prompted via schema constraints to map missing textual data to type-appropriate empty entities, specifically utilizing empty string `""` or null values for text and empty arrays `[]` for missing lists [NLL⁺26, Sat25].

During the subsequent feature engineering phase for the Random Forest classifier, these omitted values are not discarded. Instead, string-based categorical variables are transformed using Robust Ordinal Encoding. For example, the system encodes the *education* feature to a numerical scale. A missing degree or empty string is assigned a baseline weight of 0.0, whereas a PhD is represented by 4.0. This approach ensures absolute data accuracy and addresses the statistical phenomenon of Not Missing at Random data, where the absence of information is directly dependent upon the actual value of the feature itself [JSG24]. By encoding omission as the baseline ordinal state, the algorithm mathematically evaluates whether the explicit absence of credentials serves as a predictive behavioral feature in the overall candidate assessment.

3.3 Dataset Selection and Feature Engineering

3.3.1 Dataset Selection

The dataset selected for training and evaluating the proposed architecture is the “Resume and Job Description Matching Dataset” [Inc24]. It comprises 1031 resumes and corresponding job descriptions, generated and assessed using GPT-4o. The dataset provides diverse alignment scenarios, categorized into 648 unmatched, 201 mismatched and several invalid or incomplete samples. Utilizing this LLM-generated rather than real-world historical hiring data is driven by three main justifications.

Mitigation of Historical Bias and Proxy Variables

The primary motivation for excluding real-world recruitment data is the presence of historical and structural inequalities embedded within it [HMO25]. Conventional datasets frequently reflect societal disparities, meaning that algorithms processing this historical data tend to internalize and even exacerbate existing prejudices [Fab24, HMO25]. For instance, machine learning algorithms trained on historical resumes have learned to penalize female candidates by associating success with male-dominated keywords, action verbs or specific educational backgrounds [HMO25]. Furthermore, research shows that even when explicit identifiers are anonymized, real resumes remain highly vulnerable to proxy discrimination. Through the analysis of subtle language patterns, individual writing habits, and specific club memberships, algorithms can effectively deduce a candidate’s

protected characteristics, such as gender and perceived origin [BR25]. To mitigate the systemic exclusion of marginalized groups contained in human data and actively prevent the transfer of historical biases into the Random Forest, this architecture utilizes a synthetic dataset [Fab24]. Specifically, the LLM-generated data ensures a controlled, bias-reduced environment for downstream processing.

Data Privacy and Regulatory Compliance

The use of real-world resumes introduces legal and ethical complexities regarding data privacy, as these documents contain PII. The processing of PII is governed by data protection frameworks such as the GDPR and the California Consumer Privacy Act (CCPA). Transmitting and storing unmasked resumes for model training poses compliance risks and creates the potential for data breaches. Synthetic data circumvents these legal hurdles. Due to its algorithmic generation it offers a non-identifiable, privacy-preserving solution that minimizes PII exposure while still capturing the necessary statistical patterns of real world hiring dynamics, ensuring full regulatory compliance during system development [Fab24].

Controlled Environment for XAI Evaluation

Finally, synthetic datasets offer a highly flexible and controlled environment that is essential for testing XAI systems [HMO25, YAH⁺25]. By utilizing a dataset, in which the baseline scores are explicitly calculated based on predefined macro and micro criteria, researches can audit training data and improve the model’s learning capabilities without the noise and inconsistencies found in unstructured real-world data [YAH⁺25]. For the hybrid architecture proposed in this thesis, this level of control ensures that the Random Forest scoring and subsequent SHAP value calculations can be validated against known baseline scores [HMO25]. This systematic validation encourages a highly stable and transparent foundation, ensuring that the dynamic interview questions generated by the LLM are guided by objective skill assessments rather than inherited statistical biases.

3.3.2 Dataset Dropout

During the pre-processing phase, the relevance of each attribute in the synthetic dataset is evaluated before training the Random Forest classifier. The selected dataset provides structured metadata, such as pre-parsed `macro_dict` and `micro_dict` criteria, justification texts, `personal_info`, and `details` arrays. Following the relevance evaluation, these elements are excluded from the feature set. This pipeline is intentionally constrained to utilize only three components: the raw job description as plain text, the unstructured CV as plain text, and the final alignment score as the target variable.

The decision to drop all pre-processed JSON attributes is driven by the requirement to match the applicant’s profile using data independently extracted by the LLM. The extracted features are two technical skills as well as five soft skills, which are identified

in Section 3.3.4. In the context of Machine Learning, retaining redundant or irrelevant features fundamentally increases the learning difficulty of the classification model, which can lead to overfitting, reduced classification performance and decreased model interpretability. Furthermore, limiting the dimensionality of the dataset to the plain text and target score ensures optimal computational efficiency [TB24]. It forces the Random Forest classifier to focus exclusively on the derived skills, as keeping only the needed features greatly improves model interpretability and enhances predictive capabilities [IL25]. By explicitly excluding the dataset's pre-calculated metadata, the system ensures that the Random Forest and the subsequent dynamic interview questions generated remain strictly grounded in the skill extraction done by the LLM.

3.3.3 LLM-Driven Feature Engineering for Soft Skills

To generate profile-specific CV features that target soft skills and behavioral traits for distinct organizational cultures, such as the defense sector or the start-up environment, a structured prompt engineering strategy was employed. The prompt was executed using Gemini 3 Pro. The complete, unedited output generated by the model can be found in Appendix A.1.

This strategy addresses limitations of LLMs in skill extraction and facilitates the generation of explainable outputs. Aligned with recent literature, the prompt design incorporates four components: Contextual Persona, Few-Shot Learning, Domain-Specific Framing, and CoT Reasoning.

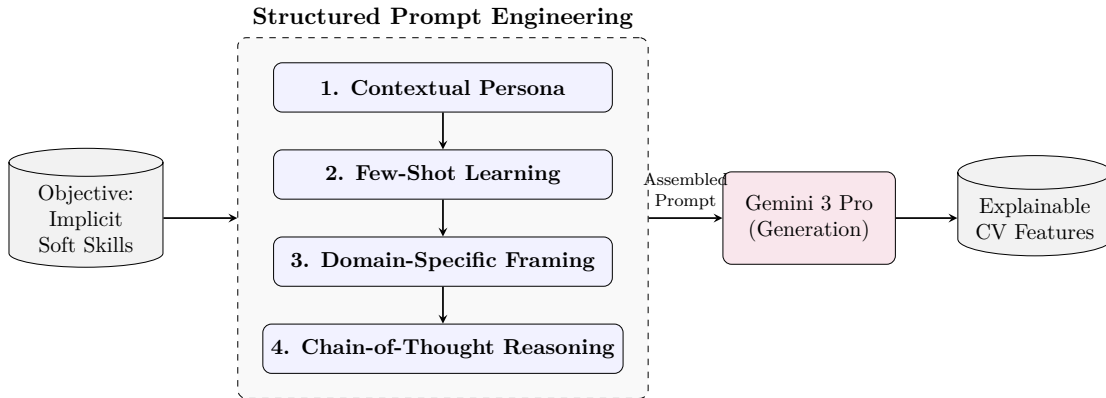


Figure 3.3: Pipeline illustrating the assembly of the structured prompt and subsequent generation of soft skill features.

Contextual Persona and Expert Framing

The prompt assigns the LLM a specific role, instructing it to act as a senior HR analyst and organizational psychologist. This technique, known as persona prompting, aligns the model with domain-specific tasks. Recent literature shows that prompting as recruitment

experts enhances their semantic alignment with HR tasks and improves their flexibility in handling unstructured text [DGK⁺25].

Few-Shot Prompting for Soft Skill Extraction

While LLMs accurately identify explicit technical skills, they often fail to infer implicit skills, such as soft skills or behavioral traits, without targeted guidance [MP25]. Consequently, the prompt includes a few-shot strategy. This approach provides the model with concrete examples of the desired output format.

Empirical comparisons of prompting strategies reveal that while some models perform adequately in zero-shot settings, the provision of structured examples significantly improves the performance of generative models in skill alignment tasks. Specifically, structured guidance helps the model bridge the gap between high-level job titles and the nuanced, non-technical competencies required for the role [MP25].

Empirical comparisons of prompting strategies reveal that providing structured examples improves the performance of generative models in skill alignment tasks. Specifically, structured guidance helps the model map high-level job titles to the non-technical competencies required for the role [MP25].

Domain-Specific Contextualization

To promote the extraction of latent features, the prompt includes different work environments. Literature on fine-tuning LLMs suggests that generic models struggle to capture domain-specific nuances without targeted data inputs. Because this architecture relies on a pre-trained model without fine-tuning, the prompt facilitates domain adaption. By embedding environmental contexts, such as “rigid hierarchy” versus “ambiguity tolerance”, the prompt aligns the generated features with the specific constraints of the organizational domain [Bha24].

Chain-of-Thought and Explainability

Finally, the prompt requires the LLM to provide a rationale for each generated feature. This aligns with the CoT principle, which is a core component of XAI in recruitment contexts. Obscuring the reasoning process of an LLM increases the risk of alignment errors and reduces decision-making capabilities. Conversely, documenting the step-by-step reasoning improves model reliability and facilitates human auditing [DGK⁺25]. Requiring the model to justify why a specific soft skill fits a given profile grounds the output in logical inference rather than stochastic hallucination. This mechanism directly reduces the black-box nature of generative models [Bha24].

Finally, the prompt requires the LLM to provide a rationale for each generated feature.

Prompt Engineering Result

I am building an XAI application for recruitment. I need to identify distinct CV features (specifically soft skills and behavioral traits) that predict success in vastly different organizational cultures.

Act as a Senior HR Analyst and Organizational Psychologist. Your goal is to identify “hidden” CV features that would make a candidate successful in specific environments.

Here is an example of the desired output:

Profile: Developer at SoundCloud

- Context: A multinational, remote-friendly culture that treats music as its DNA.
- Standard Feature: University Degree in Computer Science.
- Additional Soft-Skill Features: Inclusive Advocacy (e.g. Clouds of Color, QueerClouders, highly valued by SoundCloud), team player (low-ego approach), no-blame accountability (radical honesty to admit mistakes publicly is valued), empathy (for user-centric features), cross-functional translation (for translating technical debt into business risk and vice-versa).

Now, generate five additional CV features (focusing on soft skills/behavioral traits) for a Software Developer in the following five distinct profiles:

- Pentagon
- R&D Engineer
- Consultant
- Start-up
- Open-source Maintainer

For each feature, provide a brief rationale for why it is critical for that specific profile.

3.3.4 Generated Responses by Gemini

Using Gemini 3 Pro, the prompt generated a comprehensive list of soft skills tailor to the five distinct profiles. To integrate these identified skills into the main application’s interface as adjustable sliders, they were synthesized into five underlying soft skills:

- Structural Adherence: Defined by the degree to which a candidate thrives in established hierarchies, values protocols and respects legacy constraints. The identified soft skills pooled in this principle are executive presence, consensus building and diplomatic assertiveness.

- Adaptive Fluidity: Characterized by the capacity to function without clear instructions and well-defined tasks, switch contexts rapidly and accept undefined roles. Adaptive fluidity combines ambiguity tolerance, context switching agility, role fluidity and cross-disciplinary curiosity.
- Interpersonal Influence: Defined by the ability to navigate complex social dynamics, persuade stakeholders and drive alignment without necessarily having direct authority. This includes executive presence, consensus building and social chameleonism.
- Execution Velocity: Characterized by the prioritization of speed and output over perfection, safety or thoroughness, reflecting a bias for action rather than analysis. It pools skills such as ruthless prioritization, 80/20 pragmatism and thrift/resourcefulness.
- Psychological Resilience: Defined as the emotional stability required to handle negative input. This could include handling failure, isolation, public critique or delayed rewards. It combines emotional detachment, failure documentation, optimistic resilience and compartmentalized discretion.

3.3.5 Profile Classification Matrix

Based on the identified underlying skills, the existing five profiles were classified to construct a comprehensive evaluation matrix.

Underlying Principle	Pentagon	R&D Engineer	Consultant	Start-up	Open-Source
Structural Adherence	5	2	3	1	3
Adaptive Fluidity	1	5	5	5	2
Interpersonal Influence	2	3	5	2	5
Execution Velocity	1	2	4	5	2
Psychological Resilience	3	5	4	4	5

- Pentagon: This profile strongly contrasts the start-up profile. It requires maximal structural adherence (rated 5), because procedural rigor and hierarchical respect are non-negotiable for national security. Conversely, execution velocity and adaptive fluidity are rated lowest (rated 1), as speed and undefined roles can be harmful in this context.
- R&D Engineer: This role is defined by psychological resilience (rated 5) due to the requirement of accepting delayed rewards and failures. Given often unknown research goals, adaptive fluidity is also important.
- Consultant: Interpersonal influence is critical for this role, due to the need for executive presence and diplomatic assertiveness are necessary for consultants. Similarly, adaptive fluidity is crucial for context switching agility (rated 5).

- Start-up: The start-up profile is the only environment, in which execution velocity is rated 5. Features such as ruthless prioritization and resourcefulness highlight that speed is essential for start-ups. In contrast, structural adherence is rated lowest (rated 1), as role fluidity contradicts traditional hierarchical structures.
- Open Source Manager: This profile necessitates high interpersonal influence for consensus building and mentorship, paired with high psychological resilience to maintain emotional detachment and thick skin (both rated 5). The requirement for backward compatibility implies a lower execution velocity.

3.3.6 HR Employee Validation of Identified Features

To ensure the operational validity and real-world relevance of the LLM-generated soft skills and the subsequent profile classification matrix, expert domain feedback was solicited. LLMs can synthesize abstract behavioral traits, but these features require alignment with standard competency metrics used in talent acquisition. To evaluate this alignment, an HR professional reviewed the five synthesized principles. The review assessed whether the system’s dynamic weighting parameters reflect the specific demands of varying organizational cultures. The reviewer provided the following assessment regarding the generated skills:

I am very impressed by the selection of soft skills, which conveys a carefully considered, practical picture of the relevant competencies. In my opinion, this compilation can have a noticeably positive impact on a hiring decision. It comes across as well-founded, up-to-date, and clearly aligned with the demands of modern work environments. The only minor criticism I have concerns the area of “Interpersonal Influence”, where a candidate’s behavior in group settings could be emphasized even more and examined in greater detail. Many related aspects are already covered, but I think this area could simply be refined a bit further.

3.4 User Interface Design

3.4.1 Justification for Direct Manipulation: Likert-Scale Sliders

To facilitate the weighting mechanism, the frontend application implements Likert-scale sliders rather than traditional dropdown menus or discrete radio buttons. These sliders are anchored by qualitative labels ranging from “Ignore” to “Critical”, providing users with a cognitive map to translate their subjective hiring criteria into the system’s weighting mechanism. While conventional Likert scales are frequently implemented via discrete point-and-click buttons, recent literature emphasizes that slider scales provide a more accurate reflection of users’ true attitudes and priorities. Utilizing sliders actively mitigates common statistical phenomena such as central tendency bias, as they successfully lower the psychological tendency for users to default to explicit neutral midpoints [Cha25].

Furthermore, because sliders provide a broader and more continuous response range between two anchors, they effectively prevent the loss of measurement information and diminish the artificial ceiling or floor effects that are commonly observed in rigidly categorized discrete inputs [ZFZ⁺26].

Beyond data accuracy, the selection of sliders is heavily justified by operational efficiency and human-computer interaction principles. Sliders function as an “area-based” input method, allowing users to rapidly locate and drag the control to their desired value without the repetitive clicking required by “point-based” button interfaces [CL25]. In complex dashboard environments, stationary sliders consistently emerge as the most user-friendly interaction paradigm, demonstrating significantly lower task completion times and higher operational consistency than alternative control widgets [RAT24]. Moreover, the continuous, drag-and-drop nature of sliders supports immediate visual feedback, encouraging users to safely test reversible “what-if” scenarios, where they can immediately observe how adjustments to a feature’s importance change the candidate scores and inherent ranking. Because sliders closely align with users’ existing digital experiences and expectations, they impose a significantly lower cognitive load compared to more complex input methods [RAT24]. This ensures that a recruiter’s mental effort remains focused on evaluating candidates’ features rather than being exhausted by navigating a complex interface.

Finally, empirical evaluations of response widgets confirm that sliders yield highly positive subjective user experiences. In comparative user studies, sliders frequently outperform alternative widgets, such as dropdown menus or radio buttons, across key user experience metrics, including overall format satisfaction, perceived ease of selection and quick task completion [BR25]. Therefore, deploying Likert-scale sliders serves an efficient, transparent and user-friendly “bargaining” process between the recruiter’s subjective criteria and the system’s algorithmic baseline.

3.4.2 Model Setup

Before starting the candidate ranking, the model can be configured. By default, the API engine is automatically loaded. Alternatively, a local model can be used for document extraction and language generation.

3.4.3 Data Ingestion Interface

On startup, the interface presents a data ingestion stream as outlined in the system architecture. The system presents two distinct, side-by-side drop zones: one for the job listing and one for the candidate resumes, accepting standard document formats. This visual separation enforces the prerequisite of the system: the LLM must first process the job description and candidate CVs before candidate ranking can begin.

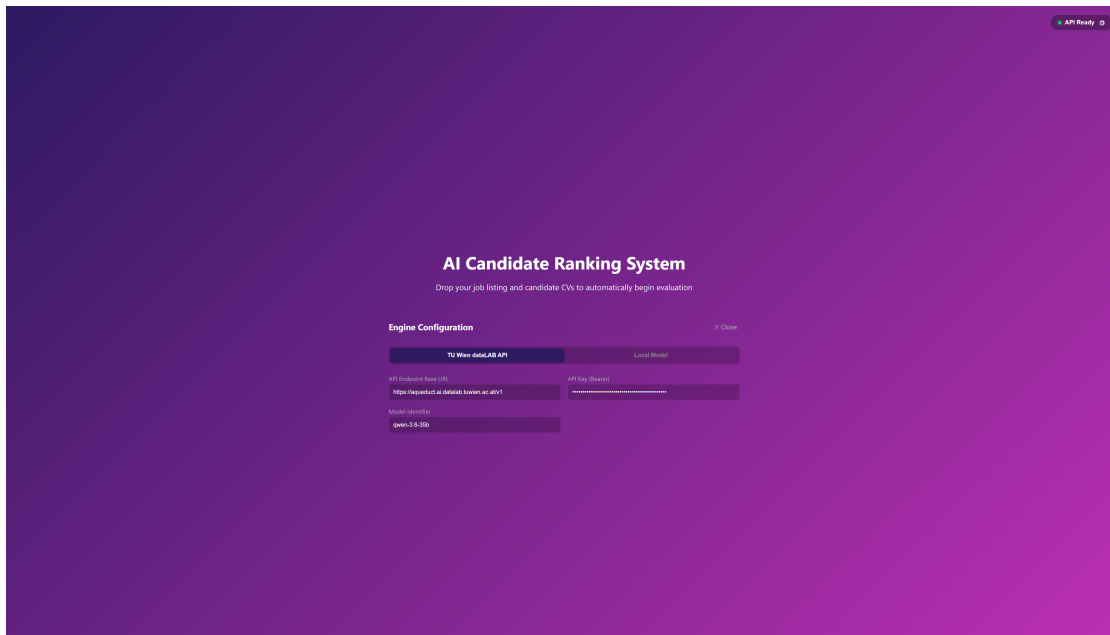


Figure 3.4: API Setup Interface

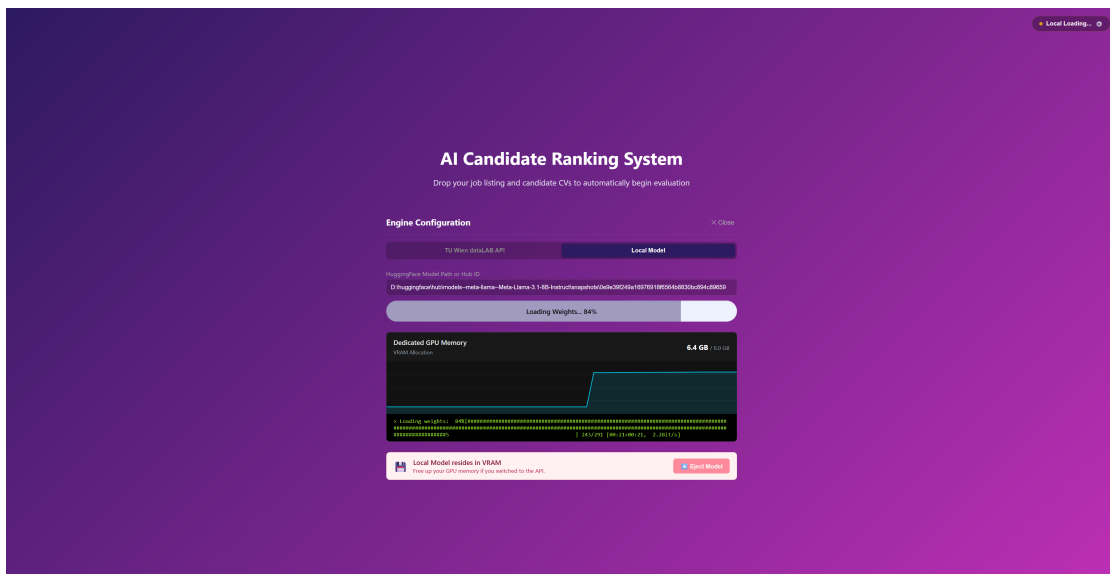


Figure 3.5: Local LLM Setup Interface

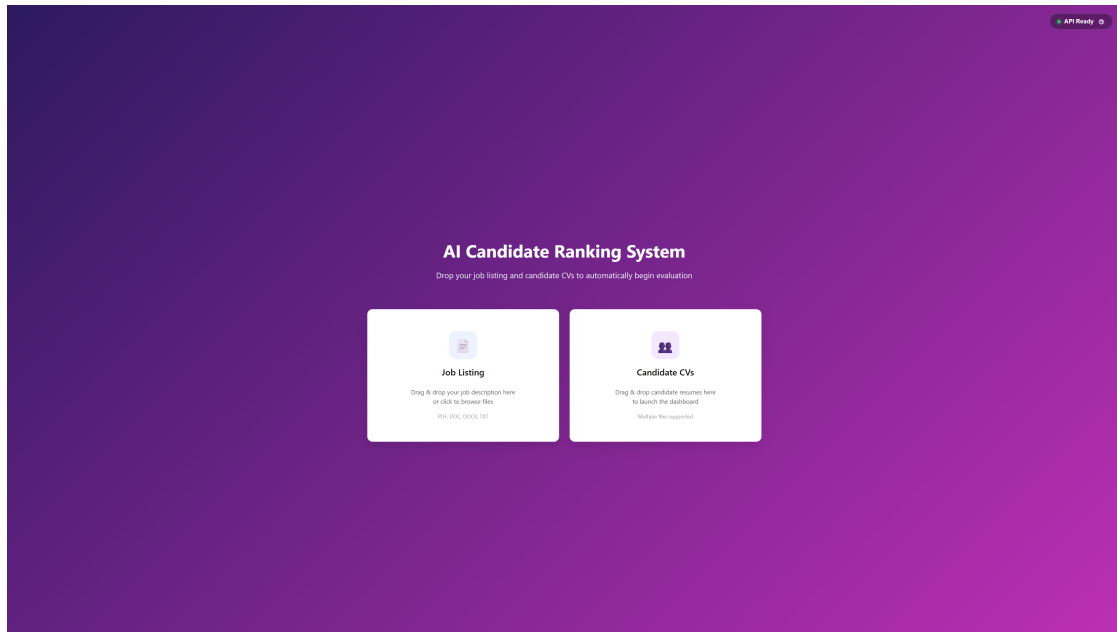


Figure 3.6: Data Ingestion Interface

3.4.4 Interactive Ranking and XAI Dashboard

After document ingestion, the system transitions to the main dashboard. The dashboard is designed to support the Human-in-the-Loop bargaining process. It is divided into two zones: the candidate feed and custom weights control panel.

The Dual-Scoring Candidate Feed

The candidate feed displays a vertically stacked list of candidates, ranked in descending order by their weighted score. To provide insights into potential discrepancies between the weighted score and baseline system score, each candidate tile features a two score bars. The top bar represents the dynamic, weighted score, while the bottom bar displays the objective Random Forest baseline score.

Dynamic Feature Weighting via Sliders

Anchored to the right side of the interface is the “Custom Weights” control panel. This section operationalizes the dynamic feature weighting through a series of interactive sliders, mapped to extracted features such as “Years of Experience”, “Education Level” and the identified soft skills in Section 3.3.4. As mentioned above, the sliders range from “Ignore” to “Critical” and the dynamic weighted score bars and the candidate ranking update dynamically.

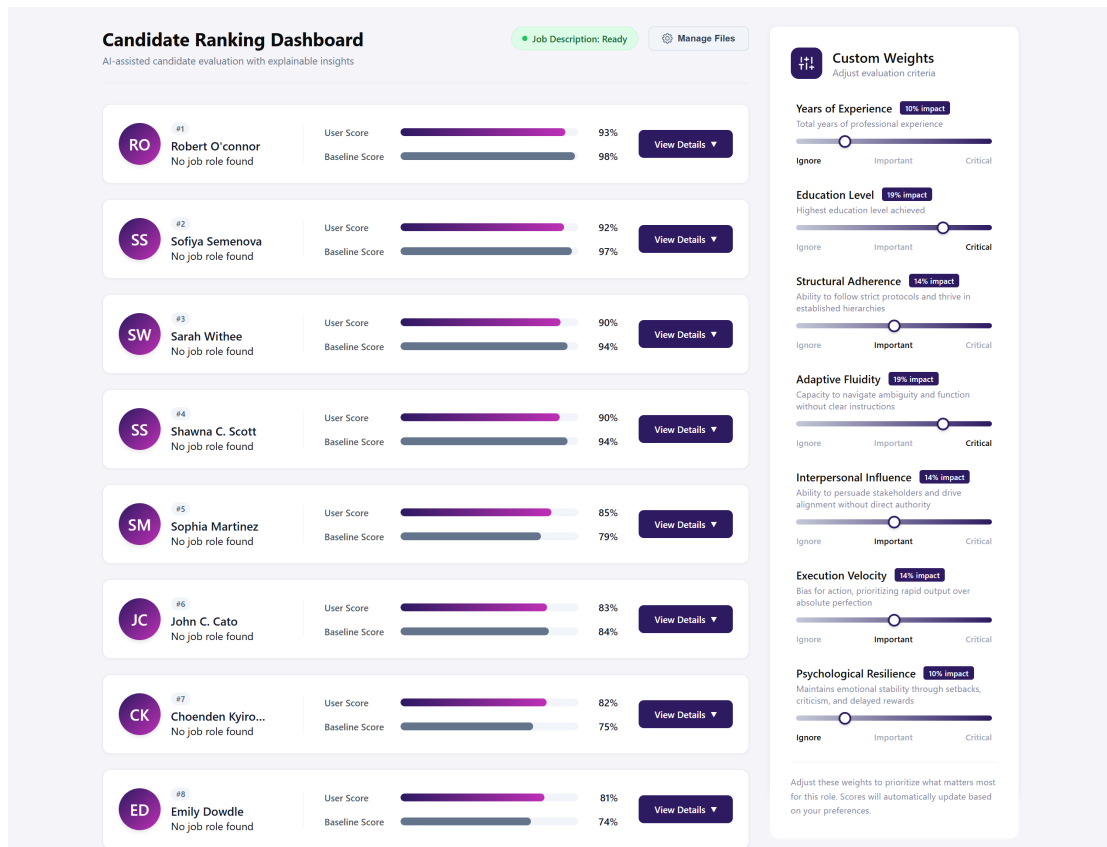


Figure 3.7: Dashboard featuring the sliders on the right and the main candidate dashboard in the middle.

Expanded Candidate Profile View

To bridge the identified actionability gap, the interface provides a detailed profile view for each candidate. When a recruiter clicks “View Details” on a specific candidate, the tile expands to reveal the generated XAI narrative. This detailed view is divided into three components:

1. **Explainable AI – Feature Impact:** A horizontal diverging bar chart, which translates the raw TreeSHAP attributions into a human-readable, visual format. Green bars represent variables that positively impacted the score, while red bars indicate negative impacts.
2. **Textual Summary:** Located in the bottom right of the expanded view, this section displays the LLM’s interpretation of SHAP values, providing a natural language description of the candidate’s evaluation.

3. Follow-up questions: Directly above the summary, the interface presents targeted, actionable interview questions generated by the LLM. Individual questions as well as all follow-up questions can be copied using a button.

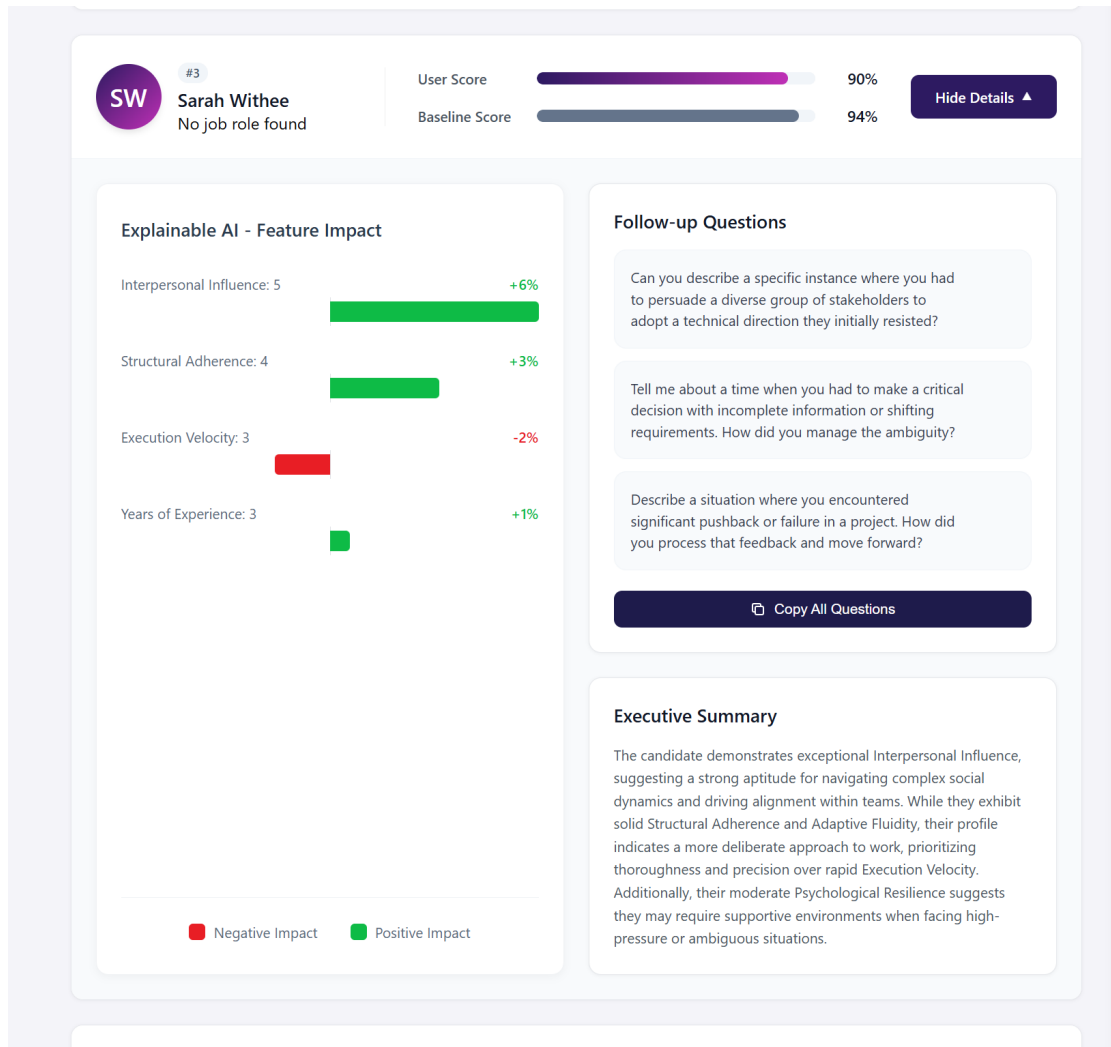


Figure 3.8: Expanded Candidate Profile View

Score Deviation Badge

If a candidate's baseline score and weighted user score differ by more than 20%, the score deviation badge is shown. On hover, a breakdown of which features contribute most to the deviation is shown. Additionally, the corresponding sliders are highlighted in the control panel.

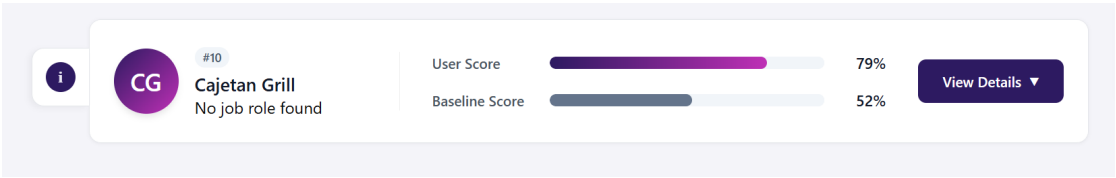


Figure 3.9: Collapsed Score Deviation Badge

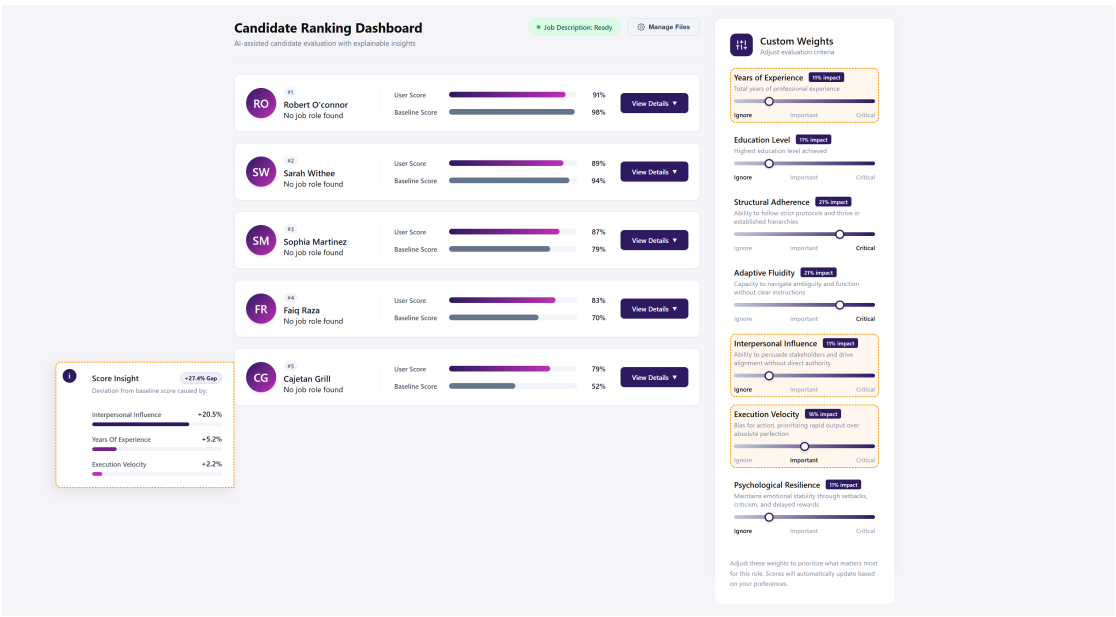


Figure 3.10: Expanded Score Deviation Badge

Implementation

This chapter details the technical realization of the recruitment pipeline described in Chapter 3. The following sections focus on the specific code-level implementations, data structures, and state management strategies used to build the asynchronous backend, the deterministic extraction modules, the algorithmic scoring engine and the reactive frontend interface.

4.1 Asynchronous Backend and Concurrency Management

To realize the event-driven architecture and prevent the user interface from freezing during high-latency LLM inferences, the backend was implemented using the FastAPI framework in Python. A central component of this implementation is the custom task management system, which relies on an `asyncio.PriorityQueue` to handle concurrent document processing and XAI narrative generation without blocking the main Asynchronous Server Gateway Interface (ASGI) event loop.

The queue is continuously polled by a background worker task initiated during the application's startup event. Tasks are pushed into the queue as tuples containing a priority integer, a task type identified, a candidate Universally Unique Identifier (UUID) and the file path. The priority system ensures that critical user interactions supersede background generation tasks:

- Priority 0 (Critical): Assigned to initial job description processing (`PROCESS_JOB`) and manual recruiter requests to prioritize a specific candidate's narrative generation (`GENERATE_XAI`).
- Priority 1 (High): Assigned to the initial CV extraction phase (`EXTRACT_CV`), ensuring that candidate profiles are parsed and visual skeleton loaders on the frontend are populated as quickly as possible.

- Priority 2 (Background): Assigned to the pre-generation of XAI summaries and interview questions, which are computationally expensive but not immediately required for the initial dashboard rendering.

All synchronous, CPU-bound operations are offloaded to separate threads using the `asyncio.to_thread()` method. These operations include Random Forest model inference, SHAP value calculations and LLM generation. The system synchronizes with the Angular frontend through a dedicated WebSocket endpoint (`/ws/candidates`), which broadcasts the current state of the active candidates dictionary and job weights at one-second intervals.

4.2 Deterministic Data Extraction Pipeline

The transformation of unstructured resume text into standardized JSON formats is handled by the `CVParseService`. A primary implementation challenge was forcing the LLM to adhere strictly to the required data schema without generating conversational padding or markdown formatting.

To achieve this, the system prompt statically embeds a strict JSON schema that maps variables like `structural_adherence` and `adaptive_fluidity` to integer types. Furthermore, the system implements the “Structural Omission” strategy discussed in Chapter 3 by explicitly instructing the LLM to output 0 for missing integers and "" for missing text fields.

Despite setting the generation temperature to 0.0 to ensure deterministic output, LLMs occasionally wrap their responses in markdown blocks, such as ````json`. To construct a robust parsing pipeline, the `_clean_and_validate` method does not rely on standard string parsing, but instead utilizes `json.JSONDecoder`’s `raw_decode` function within a loop. The algorithm scans the raw output string for the first occurrence of an opening bracket `{`, attempts to decode the object and captures the end offset. If multiple JSON objects are found, which can be the result of the LLM echoing the input schema before providing the actual data, the algorithm iterates through the parsed dictionaries in reverse, extracting the first object containing the required `Name` key. The validated data is subsequently passed into a strict Pydantic model (`CandidateFeatures`) for downstream processing.

By default, the data extraction is executed using the `Qwen3-Coder-30B-Instruct` model [Tea25, Qua25]. When the API-based engine is selected, this model is accessed via the self-hosted Aqueduct API by TU Wien’s dataLAB.

A similar deterministic extraction logic is implemented in the `JobParserService`, which evaluates the job description to establish the baseline criteria weights on a scale from 1 to 5.

4.3 Algorithmic Scoring and SHAP Integration

The core evaluation logic is encapsulated within the `ScorerService` class. During initialization, the service loads the pre-trained `.pkl` file via `joblib`. For models exported as scikit-learn pipelines, the service checks for the `steps` attribute and dynamically extracts the final Random Forest estimator. This step establishes compatibility with the SHAP explainer. Additionally, a technical override (`is_transformers_lm = lambda: False`) is applied to the SHAP library. This modification bypasses the default Hugging Face language model checks and enables the initialization of the `TreeExplainer`.

To ensure consistency between model training and live application inference, the Random Forest classifier was trained utilizing the same institutional infrastructure. The model training script was executed on an NVIDIA A40 GPU, accessed through the cluster provided by TU Wien’s dataLAB. Furthermore, the data extraction for the training set utilized the `Qwen3-Coder-30B-Instruct` model [Qua25] via dataLAB’s Aqueeduct service, directly mirroring the extraction pipeline of the deployed application.

Before scoring, the Pydantic data models are converted into a strict 1D pandas `DataFrame` via the `_prepare_feature_array` method. This process maps qualitative educational strings to numerical hierarchies, for example PhD to 4 and Bachelor to 2. The Random Forest model outputs a raw prediction between 0 and 100, which is immediately normalized to a ratio between 0.0 and 1.0.

The calculation of the dynamic user score integrates the mathematical SHAP attributions with the subjective recruiter weights. The baseline algorithmic value represents the expected value across the training data and serves as a starting accumulator. The system classifies certain attributes, such as `job_hopping`, as intrinsic features. These intrinsic features represent baseline behaviors that are not exposed to the sliders, and their raw SHAP values are added directly to the accumulator. For all other features, the slider weight (ranging from 1 to 5) is converted into a zero-centered multiplier:

$$UserMultiplier = \frac{raw_weight - 1}{4} \quad (4.1)$$

This ensures a baseline weight of 1 negates the feature’s SHAP contribution entirely, while a critical weight of 5 fully applies it. Finally, the system compares the calculated weighted score against the objective Random Forest score and shows a score deviation badge if the subjective score exceeds the objective score by a hardcoded threshold (`INFO_BOUNDARY_THRESHOLD = 0.20`).

4.4 Reactive Frontend State Management

While the backend handles heavy computation, providing real-time visual feedback during the human-AI bargaining process requires dedicated client-side state management. To

fulfill this requirement, the Angular frontend implements the `CandidateStore` using Signal-based reactive programming.

To eliminate network latency when a recruiter manipulates a feature slider, the frontend independently replicates the backend’s weighted score logic. Because SHAP is additively consistent, the frontend reverse-engineers the model’s base value dynamically. The `recalculateAndSort` function subtracts the sum of all SHAP values from the candidate’s base Random Forest score. It then recalculates the multipliers using the immediate local slider states and computes a new bound weighted score. This allows the candidate list to re-sort and trigger score deviation flags immediately within the recruiter dashboard without waiting for a WebSocket conformation from the Python server.

4.5 Context-Aware Narrative Generation

The final implementation phase translates the mathematical outputs into actionable HR insights via the `InterviewerService`. Unlike the extraction prompt which operates deterministically, narrative generation utilizes a higher temperature setting of 0.6 and active sampling to produce natural, conversational language.

The `_build_prompt` method constructs a heavily conditioned input by fusing four distinct context streams:

1. The candidate’s raw profile data serialized to JSON.
2. The raw TreeSHAP mathematical attributions representing the algorithm’s logic.
3. The recruiter’s current dynamic slider weights.
4. The status of the score deviation badge.

To ensure stable operation of the application, the prompt enforces an JSON output structure containing two keys: `executive_summary` and `interview_questions`. To safely extract this data from the generative output, the service employs a regular expression pattern (`re.search(r'\{.*\}', response_text, re.DOTALL)`) to isolate the JSON payload from any surrounding narrative text generated by the LLM before parsing it into standard Python strings and lists for the API response.

4.6 Source Code Availability

The complete source code for the fully implemented system, including the FastAPI asynchronous backend, the Angular frontend and the deterministic LLM extraction pipelines, is open-source and publicly available. The repository can be accessed on GitHub at <https://github.com/redl-m/TUW-BT> [Red26].

Evaluation

5.1 User Study Design & Procedure

To evaluate the viability of the proposed hybrid XAI framework and address the second research objective, a mixed-methods user study was conducted. The primary goal of this evaluation was to observe how users interact with the dynamic feature weighting system and to measure whether the generated textual summaries and interview questions effectively support Human-in-the-Loop decision-making.

5.1.1 Participants and Demographics

The study involved 14 participants, which registered for the user study as part of the course “193.052 Scientific Research and Writing”, within the Informatics program at TU Wien. Recent systematic reviews of XAI evaluation metrics confirm that sample sizes under 50 are highly prevalent, accounting for over 41 % of empirical XAI studies, and are perfectly justified when employing resource-intensive, real-time qualitative methods such as the Think-Aloud Protocol (TAP) and in-depth interviews [AAATAM24]. Furthermore, recruiting Informatics students serves as a highly effective “proxy user” group for this specific task [AAATAM24, NSRK24]. Their academic background ensures they possess a baseline understanding of software development roles, guaranteeing they can meaningfully comprehend the technical skills required and accurately engage with the candidate profiles.

5.1.2 Task Design and Controlled Scenario

Each individual evaluation session was designed to take approximately 45 minutes. To simulate a realistic recruitment environment, participants are placed in a fictional hiring scenario. They are instructed to act as recruiters for an early-stage start-up seeking to hire a software developer. In this role, participants upload a provided job description

and candidate CVs. Subsequently, they utilize the dashboard to screen and evaluate the applicant pool. To ensure consistency, all sessions follow a standardized protocol consisting of four distinct phases:

1. Briefing and Informed Consent (≈ 5 minutes): Participants are welcomed and informed on the study’s purpose and the fictional start-up scenario. They are provided with a participant information sheet and asked to sign a consent form, which grants permission for audio and screen recording.
2. Interactive Task and Think-Aloud Protocol (≈ 20 minutes): Participants upload the provided job description and resumes. Subsequently, they interact with the dashboard to complete the candidate evaluation task. A Think-Aloud Protocol is employed during this phase. Participants are instructed to verbalize their thought process, particularly when adjusting the sliders, examining the TreeSHAP impacts or reacting to the system’s score deviation flags. The researcher observes silently and takes timestamped notes of these interactions.
3. Standardized Questions (≈ 7 minutes): Following the completion of the task, participants fill out a digital questionnaire. This survey is designed to capture quantitative metrics regarding system usability, trust, explanations satisfaction and cognitive load. Administering the survey at this stage captures the participants’ immediate impressions of the interface.
4. Semi-Structured Debrief Interview (≈ 13 minutes): The session concludes with a qualitative interview. The researcher asks standardized questions regarding the system’s support for questioning and verifying the system’s output. Furthermore, situational questions are tailored to specific behaviors observed during the Think-Aloud phase (e.g. asking a participant why they chose to ignore a specific system warning).

To capture the qualitative data generated during the Think-Aloud and debrief phases, all sessions were audio and screen-recorded. To maintain strict compliance with GDPR regulations, the processing of audio data was strictly confined to local and institutional infrastructure, ensuring no PII is transmitted to third-party cloud services. The transcription pipelines consisted of three distinct components. First, initial audio transcription was performed using the `whisper-large` Speech-to-Text model [RKX⁺22, Sys23], which was securely hosted on TU Wien’s dataLAB infrastructure. Second, speaker diarization was executed locally using pyannote’s speaker-diarization-3.1 model [PB23, Bre23, BP24]. Finally, the raw transcripts were processed for contextual cleanup and anonymization using the `Qwen3.5-397B` model [Qwe26b, Qwe26a], accessed via the self-hosted dataLAB API. The model’s 200,000-token context window ensured session transcriptions can be processed seamlessly. The resulting transcripts and original recordings are archived within TU Wien’s infrastructure. The resulting anonymized documents were then subjected to thematic analysis. This forms the basis for the qualitative findings presented in Section 5.4.

5.1.3 Evaluation Materials and Test Resumes

To establish a realistic and controlled evaluation environment, a standardized set of nine candidate resumes was curated for the user study. Eight of these resumes were sourced from a publicly available, open-source repository dedicated to junior developer profiles [Abr16]. Utilizing this repository provided realistic CVs that accurately reflect the target demographic of the fictional hiring scenario.

To introduce necessary variance and specifically test the XAI framework’s score deviation badge functionality, a ninth resume was retrieved via a targeted search engine query.¹

Because the originally retrieved document lacked a recognized text layer, it was pre-processed using Optical Character Recognition (OCR). This step served to flatten the file and extract the underlying text, ensuring uniform machine-readability and consistent input processing across the entire evaluated corpus.

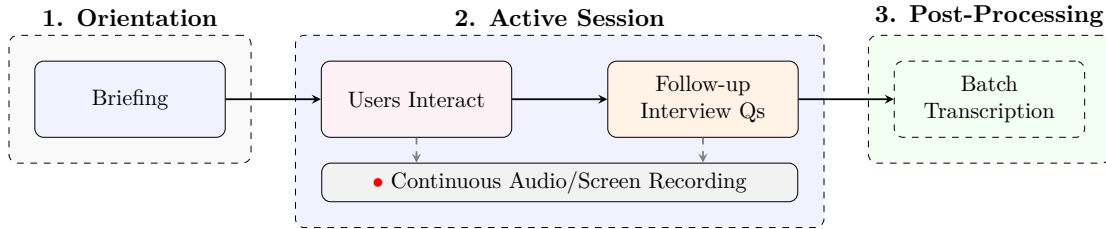


Figure 5.1: Workflow of a single user during the evaluation study, highlighting the continuous recording phase and asynchronous batch transcription.

5.2 Measurement Instruments

To comprehensively evaluate the proposed HITL XAI framework, this study employed a mixed-methods approach, integrating quantitative surveys, qualitative semi-structured interviews and real-time observational techniques. Recent literature on the evaluation of XAI emphasizes that relying solely on quantitative metrics is often insufficient, as formal evaluations may demonstrate technical accuracy but fail to capture the nuanced, subjective experiences of the end-users. The combination of quantitative data analysis and qualitative feedback facilitated an understanding of the XAI system’s performance and the cognitive processes of the users [NSRK24].

The selection of specific measurement instruments for this study was directly driven by the need to capture diverse evaluation constructs, including user trust, system understandability and perceived cognitive load, which require both objective observation and subjective self-reporting [AAATAM24].

¹The exact search string utilized was: filetype:pdf "software engineer" "resume" OR "CV" "London" OR "Berlin" OR "Stockholm" OR "Vienna"

5.2.1 Quantitative Questionnaires

Questionnaires and surveys represent the most widely used data collection method in empirical XAI research due to their efficiency in capturing a broad range of user insights in a standardized, comparable manner [AAATAM24, NSRK24]. In this study, a post-task digital questionnaire was administered immediately after the interactive phase. Utilizing quantitative scales allowed for the systematic measurement of theoretical constructs such as user satisfaction, perceived usability and confidence in the AI's recommendations [NSRK24]. Administering these standardized metrics ensures high replicability and facilitates statistical evaluation, providing an objective baseline to assess whether the dynamic feature weighting and generated narratives effectively supported the participants' decision-making processes.

5.2.2 Qualitative Semi-Structured Interviews

While quantitative surveys effectively measure how well an XAI system performs across specific metrics, qualitative research methods are essential for answering the why and exploring complex phenomena from the participants' unique perspectives. To achieve this, the evaluation concluded with a semi-structured debrief interview. The semi structured format was deliberately chosen because it balances depth with methodological consistency: It allowed the researcher to follow a standardized protocol of guiding questions while remaining flexible enough to probe unexpected behaviors or opinions that emerged during the session [Haq25]. Interviews are particularly effective for eliciting and analyzing users' mental models, allowing researchers to deeply understand how individuals perceive the AI system and interpret the provided explanations [NSRK24]. Within the context of this study, the debrief interviews provided the opportunity to ask situational questions tailored to specific behaviors observed during the interactive phase, such as exploring the exact rationale behind why a participant chose to trust the AI or ignore a specific system warning. Unlike standardized metrics, this qualitative approach enabled the documentation of personal reflections and contextual factors [Haq25].

5.2.3 Real-Time Observation: Think-Aloud Protocol

The study incorporated a Think-Aloud Protocol during the candidate evaluation phase to provide additional insights into the system usage. Unlike questionnaires and interviews, which gather retrospective feedback, TAP allows researchers to capture real-time user behavior. Using this additional data, further qualitative insights into the decision-making process and challenges encountered during system interaction can be captured [AAATAM24].

During this phase, participants were continuously encouraged to verbalize their thought processes as they navigate the dashboard, explicitly describing what they understand, what confuses them and how they assess the relevance and trustworthiness of the XAI outputs [BKK⁺25]. This continuous verbalization was crucial for uncovering the users' mental models as they are actively being formed, revealing how they interpret the

dynamic TreeSHAP impacts and react to the system’s information badges [NSRK24]. By silently observing and taking timestamped notes, the researcher can track exactly how the XAI features influence the Human-in-the-Loop decision-making process in the exact moment of interaction. Ultimately, triangulating this real-time observational data with the subjective interview responses and the objective questionnaire metrics guarantees a rigorous, holistic evaluation of the system’s true practical utility [NSRK24].

5.2.4 Questionnaire Design and Adaption

To systematically evaluate the Human-in-the-Loop XAI framework, a comprehensive 38-item digital questionnaire was constructed by adapting validated measurement scales from existing literature. The resulting instrument is divided into five logical parts designed to measure distinct theoretical constructs: explanation satisfaction, trust in automation, perceived usefulness, cognitive load and system usability. To ensure methodological consistency and ease of analysis, all 38 questions utilize a standardized 5-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree).

The theoretical constructs primarily originate from the experimental scales utilized by Wang et al. [WBC24]. These scales assess user perceptions of XAI models. Four of the five sections in the final questionnaire adapt these instruments to the recruitment and candidate screening context:

- Explanation Satisfaction (8 items): This section adapts the explanation satisfaction questionnaire by Wang et al. It measures the perceived clarity, completeness, and utility of the TreeSHAP visualizations and LLM-generated summaries. The items assess whether the explanations provide sufficient detail and clarify the model’s internal logic. Additionally, the questions evaluate if the explanations assist the user in determining when to trust the system.
- Trust in Automation (6 items): The survey utilizes the trust scale by Wang et al. to evaluate the recruiter’s confidence in the underlying Random Forest scoring mechanism. These items measure the user’s belief in the model’s competence, consistency, and dependability during candidate profile analysis.
- Perceived Usefulness (6 items): Based on the Technology Acceptance Model items used by Wang et al., this section evaluates whether the XAI features actually supported the recruiters’ decision-making and helped bridge the actionability gap. The original items were contextually modified to better reflect the system’s HR domain. For example, “learning activity” and “acquiring new knowledge” were changed to “candidate evaluation process” and “acquiring new insights about the candidates”.
- Cognitive Load (8 items): To assess the mental effort required to process the recruiter dashboard and generated narratives, the questionnaire modified Wang et

al.’s cognitive load scale. The language was adapted from a learning and question-answering context to an evaluation task, asking participants to rate the mental effort, frustration and difficulty associated with evaluating the candidates.

To evaluate general interface friction, navigation and layout of the system, such as the interactive sliders, the questionnaire incorporates the System Usability Scale (SUS) as its fifth section. The SUS is a standard measurement scale consisting of 10 items evaluated on a Likert scale, allowing for a standardized assessment of whether users found the interface unnecessarily complex, easy to use, or well-integrated [Lon18].

The final questionnaire integrates the explainability metrics from Wang et al. [WBC24] with the usability standards from Longo [Lon18]. This combined instrument measures the participants’ user experience.

5.3 Quantitative Results

This section presents the descriptive statistical findings derived from the post-task digital questionnaire. The results are structured according to the five previously established theoretical constructs.

5.3.1 System Usability (SUS)

The final section of the questionnaire utilizes the standardized 10-item SUS to evaluate interface usability and navigation. Individual item responses were processed according to the standard SUS methodology. This calculation yields a composite usability score between 0 and 100 for each participant.

The evaluation yielded a mean composite SUS score of 88.93 across all 14 participants. A detailed review of the individual usability items provides further insights. Participants agreed that the interface was easy to use ($M = 4.64$) and that they imagined most people would learn to use it very quickly ($M = 4.86$). When asked if they would need the support of a technical person to use the interface, every participant strongly disagreed, yielding a minimum mean score of 1.00. Furthermore, the respondents reported feeling confident while using the interface ($M = 4.14$) and indicated that the interactive functions were well integrated ($M = 4.43$).

5.3.2 Trust and Explanation Satisfaction

The items measuring explanation satisfaction focused on the clarity and utility of the LLM-generated summaries and TreeSHAP visualizations. Participants indicated that the explanations clearly told them how to use the model ($M = 4.64$) and that the insights were useful for their evaluation goals ($M = 4.36$). However, these scores were lower when participants were asked if the explanation showed how accurate the model is ($M = 3.36$) or if the explanations let them judge when to trust or not trust the model ($M = 3.29$).

Within the trust in automation construct, participants generally agreed that the AI model is a competent performer ($M = 3.57$) and that they can rely on it to behave in consistent ways ($M = 3.79$). Conversely, when responding to the direct statement “I trust the model”, the average response dropped below the neutral midpoint to 2.86.

5.3.3 Perceived Usefulness & Cognitive Load

The perceived usefulness metrics capture the participants’ assessment of the framework’s impact on their recruitment task. Participants strongly agreed that the AI model was helpful in acquiring new insights about the candidates ($M = 4.21$) and that the features smoothed the overall screening process ($M = 4.07$). Responses were comparatively lower regarding the statement that the AI-embedded approach is more useful than conventional manual CV screening ($M = 3.14$).

The cognitive load scores remained consistently low across all observed metrics. Participants largely disagreed with the notion that they had to put a lot of effort into evaluating the candidates ($M = 2.14$) or that the workflow caused them a lot of mental effort ($M = 1.64$). Furthermore, users reported low frustration during the activity ($M = 1.79$) and did not find the information present in the interface to be difficult to understand ($M = 1.64$).

5.3.4 Task Completion and Candidate Selection Metrics

In addition to the subjective questionnaire data, the evaluation recorded objective behavioral metrics during the interactive screening task, specifically focusing on the time required to complete the candidate selection and the final hiring decisions made by the participants.

Regarding task efficiency, the 14 participants required an average of 16.29 minutes to complete the candidate screening and selection process. The completion times exhibited significant variance across the study group. The fastest participant concluded the task in 3 minutes, while the longest session required 39 minutes to arrive at a final decision.

In terms of candidate selection, participants were tasked with choosing their top two applicants from the provided pool. The behavioral data reveals a strong, but not absolute adherence to the system’s algorithmic rankings. In the majority (64.3%) of the sessions, participants selected the candidates explicitly ranked as #1 and #2 by the system. In 21.4% of sessions, the top voted candidate was chosen alongside a lower rated candidate ranked #3 or lower. In the remaining 14.3% of sessions, users bypassed the #1 ranked candidate entirely and selected the candidates ranked #2 and #3.

This shows that in all of the sessions, one of the two chosen candidates was induced in the best two candidates as ranked by the system. In the majority of all cases, the candidates that have been rated best by the system were the ones to be chosen. In all other sessions, at least one of the two candidates chosen was ranked #1 or #2 by the system.

5.4 Qualitative Findings

This section presents the qualitative findings derived from the transcribed Think-Aloud protocols and semi-structured debrief interviews conducted during the user study. The observations are structured categorically around central themes, documenting participants’ interactions, recurring points of confusion, general integration workflows and specific feature requests.

Thematic Category	Key Observation / User Behavior	Frequency ($N = 14$)
Slider Interaction & “Bargaining”	Proactively set sliders before reviewing candidate details to avoid baseline anchoring.	6
	Relied on system defaults as a helpful starting point or left them completely unchanged.	5
	Viewed sliders as an essential tool to exert influence and input personal preferences.	7
Trust & Explainability	Reported difficulty reconciling negative SHAP values with highly rated soft skills.	7
	Questioned the capability of the LLM to infer complex soft skills.	7
	Requested a traceability feature connecting inferred skills to the original CV text.	6
Narratives & Questions	Praised the structural consistency of summaries for preventing “glossing over” details.	4
	Viewed generated interview questions as valuable “blueprints” and inspiration.	10
Workflow & UI Elements	Emphasized the persistent necessity to view the original, unedited CV alongside the AI.	13
	Misunderstood the initial purpose or meaning of the Score Deviation Badge.	3

Table 5.1: Summary of key qualitative findings from the Think-Aloud protocols and debrief interviews, categorized by central themes and observed frequencies.

5.4.1 The “Bargaining” Process: Interactive Sliders and Custom Weights

Participants interacted with the custom sliders to apply distinct strategies to configure their baseline preferences. Seven users viewed the sliders as an essential tool to exert influence over the algorithmic outcomes. One participant noted, “[...] without it, it would be like just the AI rating people, not really giving me any chance to give it like an input what I’m really searching for [...]”.

Six participants proactively set the sliders before looking at the rest of the system, with one user mentioning that they wanted to avoid being biased by the system’s baseline. A user explained their workflow: “[...] it has done a good job prioritizing these scores, but still I think like [...] that [...] it’s important that you do that [adjusting the sliders] first and then afterwards you just check if your inside like evaluates the same things as what the AI or whatever does”. Another participant mentioned having “one specific slider setting in mind”, but expanded some candidate’s profiles before adjusting the sliders.

Another participant completed their slider configurations entirely upfront: “So first, I had my view of how I want this perfect candidate to be, and so I like build it with these parameters and I didn’t come back [...]”. Conversely, three viewed the system presets as helpful baselines, stating “No, they were a good starting point. I would also say that if I would use this tool more often, I would rely more on them at the beginning of an evaluation”. Two participants did not change the pre-adjusted slider settings at all and went with the default settings.

5.4.2 Trust, Transparency and the Feature Impact Diagram

While the feature impact diagram was heavily utilized to gain an overview of a candidate, it triggered points of confusion for ten candidates, primarily concerning the color-coding and score baseline representations: Out of these ten participants, seven struggled to comprehend why highly-rated soft skills (e.g. scored 5/5) were displayed with red bars indicating a negative impact on the overall score.

One participant highlighted this issue: “Even though there’s psychological resilience, but I think all candidates are in the red one for that. I’m not quite sure yet what that means exactly, because I don’t know what minus one percent means for the AI”. This sentiment was shared by seven participants in total, who expressed similar confusion: “I want to say that the rating in combination with the the color scheme is a little bit confusing if you don’t know how, what is weighted and how it is scored”. The visual discrepancy caused notable friction: “[...] it’s a little bit confusing with the impact and the different... [items listed] [...]. [...] I would need more information about how the AI works and how it ranks because for me now it’s a [...] cognitive dissonance in my head when I see that [scores rated the maximum number having a negative impact on the overall score]”.

Furthermore, seven participants questioned how the LLM accurately derived complex soft skills solely from a standard CV. One participant remarked, “[...] for example psychological resilience is a thing where I’m thinking that maybe that’s hard to kind of extract from a CV even as a person because how would I know how resilient a person is without having experienced them in a really psychological hard environment. So I’m kind of critical of these”. Another stated, “I’m not sure how I would know that. Or how AI would know if someone follows strict protocols based on an application or CV.” Implementing a traceback feature directly linking to CV citations was suggested or

confirmed by six participants, with one user noting “Yes, that would be great. Then I think I would trust that [the feature impact diagram] more”.

5.4.3 The Executive Summary and Interview Questions

The textual executive summaries and interview questions were positively received, serving as efficient tools for candidate comparison and evaluation.

One participant observed that the summaries humanized the applicants, stating, “I think the summary is very nice because it kind of makes the person feel more like a person [...]”. Furthermore, four participants appreciated the structured nature of the summaries for comparison: “[...] a good thing to have maybe is that you basically read the same thing for everybody in the sense that you might, if you read 50 CVs yourself, you might gloss over some parts [...] and you give basically the same chance to everybody”. However, one participant noted a potential bias in the tone: “I think this executive summary is a little bit, I want to say, bias from the AI so that there is an opinion in there of the AI and I think it’s biasing me too when I read it because when I get lower the candidates get described more bad and when I’m on the higher rank other candidates are described more good”. Following up this statement, the participant explained that “I’m not sure if it’s highlighting the negative things or only highlighting that this person is not right for this job description”.

The interview questions were regarded as valuable inspiration and a baseline for interviews by ten participants. A participant explained, “basically it definitely helped me find some new perspectives [...] I think it’s also nice for experienced people to see because it’s it’s always good to have some inspiration [...]”. Another mentioned, “[...] if I feel a need to ask something like that I have like so my blueprint”.

5.4.4 General Integration, Workflow and the Need for Resumes

The recurring necessity to view the actual CVs despite the AI’s breakdown was a dominant finding, explicitly expressed by thirteen participants. Furthermore, the system was viewed by six participants as a first step, “add-on” or second opinion to the traditional recruitment pipeline.

One user summarized this sentiment: “[...] this summary is very short and I wanted to see the whole CVs and the people behind it. And I wanted to make my own picture of it because I, I don’t know if I’m comfortable with saying no to people only based on this executive, executive summary or at a future impact or something”. The same user highlighted how the system scales with volume: “Normally I would say if I have like, I don’t know, 10 people, I would first read the CVs and then use the system to make sure my opinion is the same or to confirm it or something. And if I would have, I don’t know, 100 people wanting to work in my firm and don’t have time to look at all the CVs, then I would use the system first. Then I would use the slides on the side”.

The score deviation badge, intended as a way of highlighting a candidate being heavily boosted by a recruiter's current slider settings, was misunderstood by three participants. Responses included, "[...] if I'm being honest I didn't really to 100 % understand what it meant so I didn't know what I should do with this information on that board" and "And I didn't really understand straight away what this little sign means". After getting further clarification about the badge, three participants agreed on the badge being useful: "Yeah, this makes sense very much. And I think if I understood that before, I would have thought differently about this person.". Additionally, as the badge was shown for candidate ranked #8 by default, the badge did not influence two users' judgment as the candidate was rated low anyways: "The candidate wasn't even on my radar because his rating was so low [...]".

5.4.5 Recurring Requests and Suggestions for Improvement

Throughout the sessions, participants voiced specific requests to improve system usability and comparative analysis:

- Side-by-Side Comparison: Two Participants wanted to compare candidates directly. "Okay, I think if I am comparing two candidates, I would have preferred it to have them both, yeah, one beside another, because by having it like this, I have to scroll, and having it side by side, it's easier to compare them."
- Standardized Feature Impact Diagram: Six users found it difficult to compare candidates using the feature impact diagrams, as it showed different features for different candidates: "It's difficult to compare the candidates when they don't have the same features". They suggested having multiple viewing options, where users can pick the features they want to see using checkboxes.
- Hover Explanations/Tooltips: To aid the understanding of complex soft skills, four participants requested hoverable pop-ups. "You could have like a pop-up window here. If you navigate here and your mouse is here and it kind of opens up a bigger description of what each weight means, maybe that would be better".
- Hard Skill Checkmarks: Two Users requested visible checklists for basic requirements. "[...] you could like add some check marks or something where you can check like the abilities for example here you need dart or [...] python or whatever so that it takes that also as like a score".

Discussion

The overarching objective of this thesis was to investigate and resolve the “actionability gap” inherent in current XAI recruitment systems. By designing, implementing and evaluating a hybrid architecture that integrates the structured transparency of Random Forests with the generative capabilities of LLM, this research aimed to shift XAI from a passive analytical output to an active, human-centric decision support tool.

While the evaluation chapter detailed the specific quantitative measurements and qualitative observations collected during the user study, this chapter synthesizes those findings to interpret their broader implications. It puts the proposed framework into perspective, contrasts the empirical results with existing literature and acknowledges the limitations of the current study.

6.1 Empowering the User: The “Bargaining” Process

A fundamental socio-technical requirement for modern HR systems is the ability for users to “bargain” with the system. This means, users can negotiate, contextualize and adapt the algorithm’s outcomes to match their shifting, subjective organizational criteria [BBF⁺23, PMG25]. The proposed system operationalizes this requirement by extending passive XAI outputs with dynamic Likert-scale sliders, answering the academic call for interactive bias mitigation.

The evaluation overwhelmingly confirms that this interactive mechanism significantly enhanced the users’ perceived control. The ability to manipulate the algorithmic baseline allowed participants to transition from passive consumers of AI rankings to active participants in the evaluation pipeline. Interestingly, the TAP data revealed that six users deliberately adjusted the sliders before reviewing the AI’s recommendations. This proactive strategy was employed to actively prevent anchoring bias, ensuring that the user’s ideal candidate profile was established independently of the LLM’s preset weights.

This interactive feedback loop proves that integrating dynamic feature weighting successfully adapts to evolving evaluation criteria. By placing the human strictly in the loop, the system ensures that the final candidate ranking is a collaborative synthesis of the Random Forest’s objective data processing and the recruiter’s subjective priorities.

6.2 Bridging the Actionability Gap

The most significant architectural contribution of this thesis was the attempt to help bridge the actionability gap by translating raw SHAP values into dynamic, LLM-generated executive summaries and targeted interview questions. Existing literature has frequently criticized traditional XAI outputs for providing mathematical transparency without offering practical utility for the end-user’s actual tasks [NSRK24].

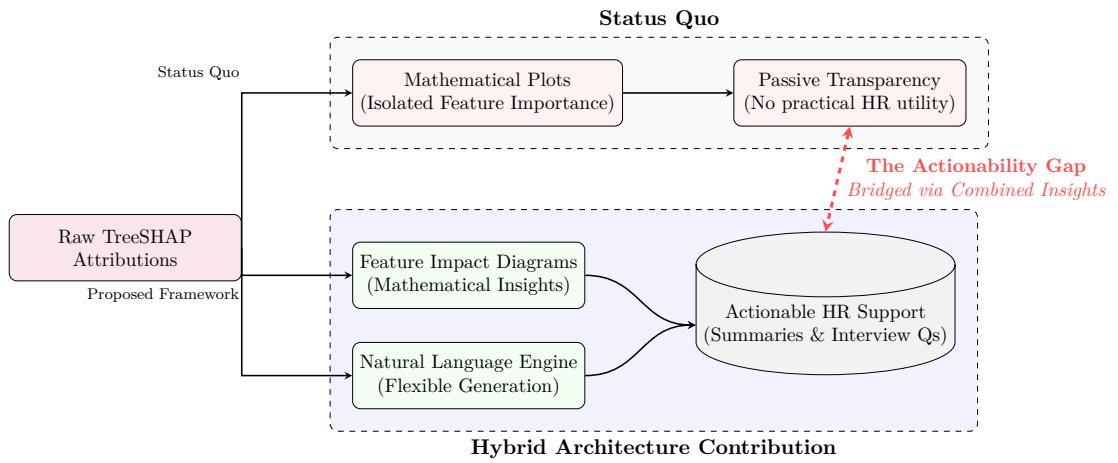


Figure 6.1: Visual comparison of traditional XAI outputs versus the proposed hybrid architecture, illustrating how combining mathematical impact diagrams with generative natural language bridges the actionability gap for HR professionals.

6.2.1 Standardization and Candidate Humanization

The introduction of natural language executive summaries proved highly effective in reducing the cognitive load of the screening process. Participants praised the summaries for providing structural consistency, noting that reading a standardized narrative for every candidate prevented them from subconsciously “glossing over” important details – a common pitfall when manually reviewing dozens of uniquely formatted CVs. By synthesizing disparate data points into a cohesive narrative, the LLM effectively “humanized” the applicants. This findings aligns with recent studies indicating that narrative-based explanations significantly improve user comprehension and satisfaction compared to raw mathematical plots [NBG⁺25].

However, the qualitative feedback also surfaced a potential risk: generative bias in tone. Two participants noted that the LLM’s language felt inherently more negative when

summarizing lower-ranked candidates, potentially biasing the recruiter’s perception before they even read the details. This underscores the necessity for strict prompt engineering to enforce a strictly neutral, objective tone in generated summaries, regardless of the candidate’s algorithmic score.

The evaluation revealed that standardization must extend to the interface’s visual components to support comparative analysis. Qualitative feedback indicated that users experienced difficulty comparing candidates when feature impact diagrams displayed different metrics. Participants reported that altering the displayed features based on local importance induced cognitive dissonance. Therefore, a core design principle emerges: Ensure standardized visual comparisons. To facilitate fair candidate comparison, feature impact diagrams must display a consistent set of metrics across all applicants and systems should allow users to toggle between a standardized global view and a candidate-specific local view.

6.2.2 Interview Questions as Strategic Blueprints

The generation of actionable interview questions represented the final step in moving the system from a passive analytical tool to an active recruitment assistant. The response to this feature was universally positive. Participants viewed the questions as highly valuable “conversation starters” that successfully identified new evaluation angles they might not have considered independently.

Crucially, users did not view these questions as rigid scripts. The qualitative data reveals that recruiters intend to use the AI-generated questions as a strategic baseline or blueprint, adapting and rephrasing them to fit the natural flow of a human conversation. This perfectly encapsulates the goal of the hybrid architecture: the AI does not dictate the human’s actions, it supports their capabilities. Providing inspiration to investigate specific TreeSHAP impacts enables the system to bridge the actionability gap, allowing recruiters to actively verify the system’s findings.

6.3 The Persistent Need for Primary Document Verification

Despite the high SUS scores and the proven effectiveness of the dynamic dashboard, a recurring theme throughout the user study was the participants’ persistent desire to review the original candidate CVs. Participants emphasized that they would not feel comfortable making a final rejection or hiring decision based solely on the AI’s executive summary and feature impact diagrams.

This finding is highly significant for the deployment of generative AI in HR. It confirms that even with advanced, highly transparent and interactive XAI narratives, AI systems are fundamentally viewed as preliminary pre-selection tools, not autonomous decision-makers. Users emphasized taking original formatting, specific project descriptions and the overall effort demonstrated in a CV into account. Therefore, an optimal HR system

must not attempt to abstract the CV entirely. Instead, the AI dashboard should serve as a comparison view, presented side-by-side with the original document. This leads to two additional design directives for HR interfaces.

First, developers must maintain primary document access. Actionable XAI should complement, not replace, the source material. Dashboards should utilize a side-by-side UI layout, displaying the AI’s XAI narratives immediately adjacent to the original applicant CV, allowing the AI dashboard to serve as a direct comparison view.

Second, to satisfy the need for immediate factual verification alongside the original document, systems must integrate hard skill checkmarks. While XAI visualizes complex, weighted scores, recruiters still require verification of absolute prerequisites. Combining abstract feature impact diagrams with simple, verifiable checkmarks drastically reduces cognitive load during document verification.

6.4 From Absolute Trust to Calibrated Reliance

A central premise of early XAI research was that increased transparency would automatically lead to increased user trust, which was viewed as the ultimate metric of success [Fab25]. However, recent literature on human-machine teaming argues that striving for absolute trust is fundamentally flawed. Instead, the goal of a transparent system must be to foster “calibrated trust”. Trust is considered properly calibrated when a user’s reliance on the system accurately matches the system’s actual trustworthiness and competency in varying situations. Misalignment leads to either undertrust, which is the disuse of a capable system, or overtrust, also known as blind automation bias [MB19].

The empirical findings of this thesis demonstrate that the proposed hybrid framework successfully promotes calibrated trust rather than blind automation bias. As observed in the qualitative data, participants recognized the AI as a competent and consistent assistant, yet the direct measurement of absolute trust fell below the neutral midpoint. Rather than indicating a failure of explanations, interpreting this metric alongside the qualitative TAP data reveals a successful Human-in-the-Loop dynamic. The system actively prevented the “illusion of fairness” [Fab25]. Users did not treat the AI as a source of absolute trust, but retained a healthy, analytical skepticism.

This calibrated reliance is most visible in the candidate selection behavior. While the majority of the sessions resulted in the selection of the algorithmically top-ranked candidates, over a third of the interactions involved active deviations. By utilizing the generated summaries and their own dynamic feature weights, participants bypassed the highest-ranked options when the AI’s logic did not align with their contextual understanding of the fictional start-up scenario. The transparency of the TreeSHAP visualizations empowered users to selectively rely on the automation only when they deemed it appropriate, directly fulfilling the core requirement of calibrated reliance [MB19, NBG⁺25].

6.4.1 The “Secondary Black Box” and the Limits of LLM Inferences

Despite the success of the overarching architecture in fostering calibrated trust, the qualitative synthesis revealed a critical boundary regarding the transparency of LLM-driven feature engineering. While participants easily trusted the evaluation of explicit metrics, such as “Years of Experience”, there was widespread skepticism regarding how an AI could accurately derive complex behavioral traits, such as “Psychological Resilience”, purely from the unstructured text of a CV.

As noted by multiple participants, assessing psychological stability or interpersonal influence is challenging even for humans during face-to-face interviews. Consequently, presenting these inferred soft skills as definitive numerical scores without displayed the underlying textual evidence inadvertently created a “secondary black box”. The TreeSHAP diagrams successfully explained how the Random Forest weighted the soft skills, but they failed to explain why the LLM assigned those soft skill scores in the first place. To fully calibrate user trust and counter algorithmic aversion, future iterations of the system must implement strict data traceability. Systems must not present LLM-inferred soft skills as absolute facts. Instead, any behavioral trait or soft skill extracted by an AI must feature a traceback mechanism linking directly to exact citations from the original CV that justifies the inference. Directly linking these extracted feature to the source material would bridge the gap between generative AI inference and human verification.

6.4.2 Calibration Points and UI Salience

In calibrated trust theory, a “Calibration Point” is a specific situation where the automation’s performance conditions change meaningfully, requiring the user to adjust their trust accordingly [MB19]. The score deviation badge serves as such point, warning recruiters when a candidate’s user score was being artificially pushed due to current slider settings rather than algorithmic scoring.

The qualitative synthesis demonstrates that while the theoretical implementation of this calibration point was sound, its UI execution was insufficient. Participants who understood the badge found it highly valuable and stated it would have changed their perspective on a candidate. However, a significant portion of the user base either ignored the badge or misunderstood its meaning. This highlights a crucial lesson is XAI design: developers must design salient calibration points. As the user study has shown that subtle notifications are frequently ignored, notifications about system confidence or score deviations must be highlighted to modulate user trust. Furthermore, notifications acting as calibration points must provide explanations for the system’s behavior to calibrate reliance [MB19]. This could for example be a textual explanation of why the system flags a discrepancy.

6.5 Contextualizing User Behavior and Future Prospects

It is crucial to acknowledge that the user behaviors observed during this evaluation represent a temporary “snapshot” of initial interactions rather than established habits. Because participants were placed in an artificial, controlled study scenario, their engagement with the interface naturally reflects first-time usage. The qualitative feedback strongly suggests that these interaction pattern would evolve with prolonged system exposure.

For instance, while six users proactively adjusted the slider weights to establish their own baseline and avoid anchoring bias, three participants indicated that increased familiarity with the dashboard would alter this approach. Reflecting on the system’s baseline weights, one participant noted: “I would also say that if I would use this tool more often, I would rely more on them at the beginning of an evaluation. [...] I think they [the slider presets] were pretty spot on.” This highlights a critical dynamic: as users become accustomed to the interface and their trust calibrates over time, their “bargaining” behavior might shift from active manual adjustments to a heavier reliance on algorithmic defaults. Consequently, observing how non-technical HR professionals interact with the introduced system during their actual, day-to-day work routines over an extended period remains a vital necessity for future research. Longitudinal field studies in real-world work situations are required to fully validate the system’s long-term operational impact.

Finally, to support this long-term integration, developing advanced technical features to further reduce the “secondary black box”, for example using Retrieval-Augmented Generation, must remain a priority. Specifically, realizing the LLM-inference citations traceback stands out as the most prominent prospect for future technical iterations, allowing to check the inferred soft skills directly with the linked, textual evidence from the original resumes.

6.6 Prototype Adaptions

To bridge the gap between the qualitative evaluation findings discussed throughout this chapter, a series of interface iterations were conceptualized using a Figma prototype. This phase visualizes the direct implementation of the participants’ most prominent feedback.

To provide a clear baseline and visualize the changes, these adaptations are shown side-by-side with the current state of the application.

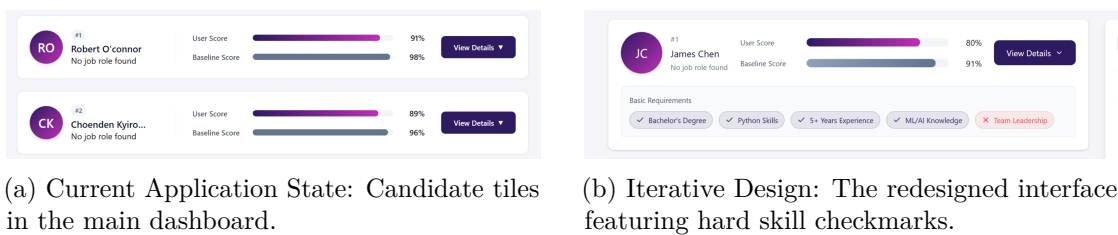


Figure 6.2: Overview of the transition from the evaluation phase to the iterative redesign.

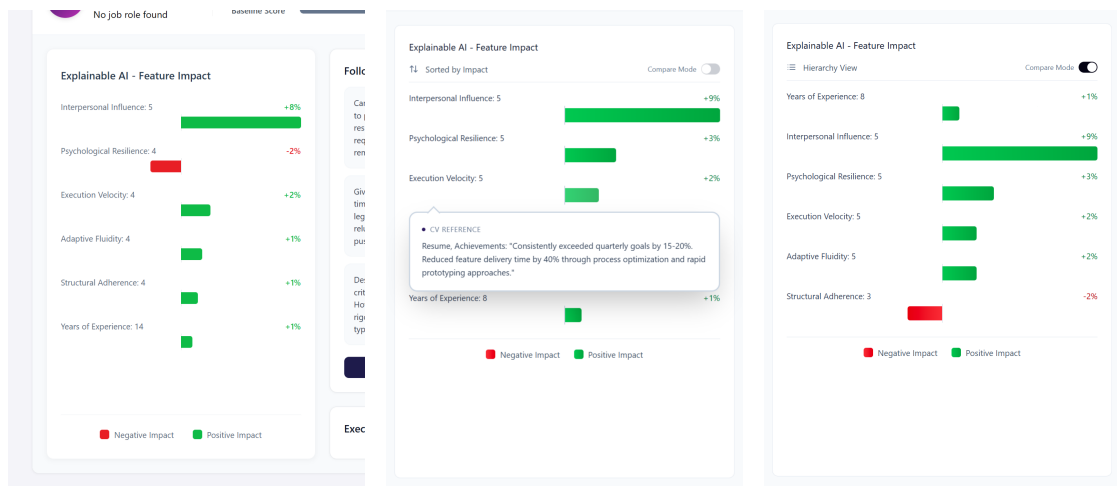
Hard Skill Verification: Visible checkmarks were integrated directly into the candidate dashboard to facilitate quicker baseline filtering. This feature reduces reliance on the primary document and enables recruiters to verify mandatory technical requirements.

Contextual Soft Skill Explanations: Hover-triggered tooltips were implemented to reduce the reported cognitive dissonance regarding the scoring and weighting of complex attributes. These tooltips provide standardized definitions of each soft skill and explain its influence on the AI's evaluation.



Figure 6.3: Contextual soft skill tooltip.

6. DISCUSSION



(a) Current State: Isolated feature impact diagram. (b) Iterative Design: Soft skill traceback to CV. (c) Iterative Design: Comparison mode.

Figure 6.4: Addressing the secondary black box and comparison friction through interface redesign.

CV Traceability Mechanism: To reduce the “secondary black box” effect of generative AI inferences, a traceback feature was introduced. When reviewing a candidate’s soft skills, users can now follow a direct link to the exact citation or section within the uploaded resume that prompted the model’s specific rating, anchoring the AI’s output in the ground truth.

Standardized Comparison Mode: Solving the friction of comparing candidates with disjointed feature impact diagrams, the prototype introduces a comparison mode. This view aligns candidates along a unified, user-selected set of features, enabling direct visual contrast of their respective strengths, weaknesses, and algorithmic penalties.

Design Guidelines for Actionable XAI in Recruitment

As derived from the empirical findings and discussed in the previous chapter, bridging the actionability gap requires a shift from passive transparency to active, Human-in-the-Loop system design. While Chapter 6 explored the theoretical implications of these findings, this chapter extracts and sharpens those insights into a concrete set of design guidelines for future developers implementing actionable XAI tools in HR.

These guidelines are not intended as universal design principles for all AI applications. Instead, they serve as targeted interventions to mitigate the socio-technical friction that arises when non-technical HR professionals interact with complex algorithmic outputs [SVG⁺25]. Consequently, the successful deployment of actionable XAI requires that systems:

- **Maintain Primary Document Access:** Actionable XAI should complement, not replace, the source material using side-by-side UI layouts that display the AI's narratives immediately adjacent to the original applicant CV.
- **Integrate Hard-Skill Checkmarks:** While XAI excels at visualizing complex, weighted scores, recruiters still require immediate verification of absolute prerequisites (e.g. "Has a Bachelor's degree", "Knows Python"). Combining abstract feature impact diagrams with simple, verifiable checkmarks drastically reduces cognitive load.
- **Ensure Standardized Visual Comparisons:** Feature impact diagrams must display a consistent set of metrics across all candidates to support objective evaluation. Furthermore, the interface must enable users to toggle between a standardized global view and a candidate-specific local view.

7. DESIGN GUIDELINES FOR ACTIONABLE XAI IN RECRUITMENT

- **Implement Data Traceability:** Traceability mechanisms must link LLM-inferred behavioral traits to the underlying textual evidence. This approach prevents the presentation of inferred soft skills as verified facts.
- **Design Salient Calibration Points:** Ensure that visual notifications regarding system confidence or score deviations are highly prominent, as subtle icons are frequently ignored. Calibration points must explicitly state why the system is flagging a discrepancy to accurately calibrate user trust [MB19].

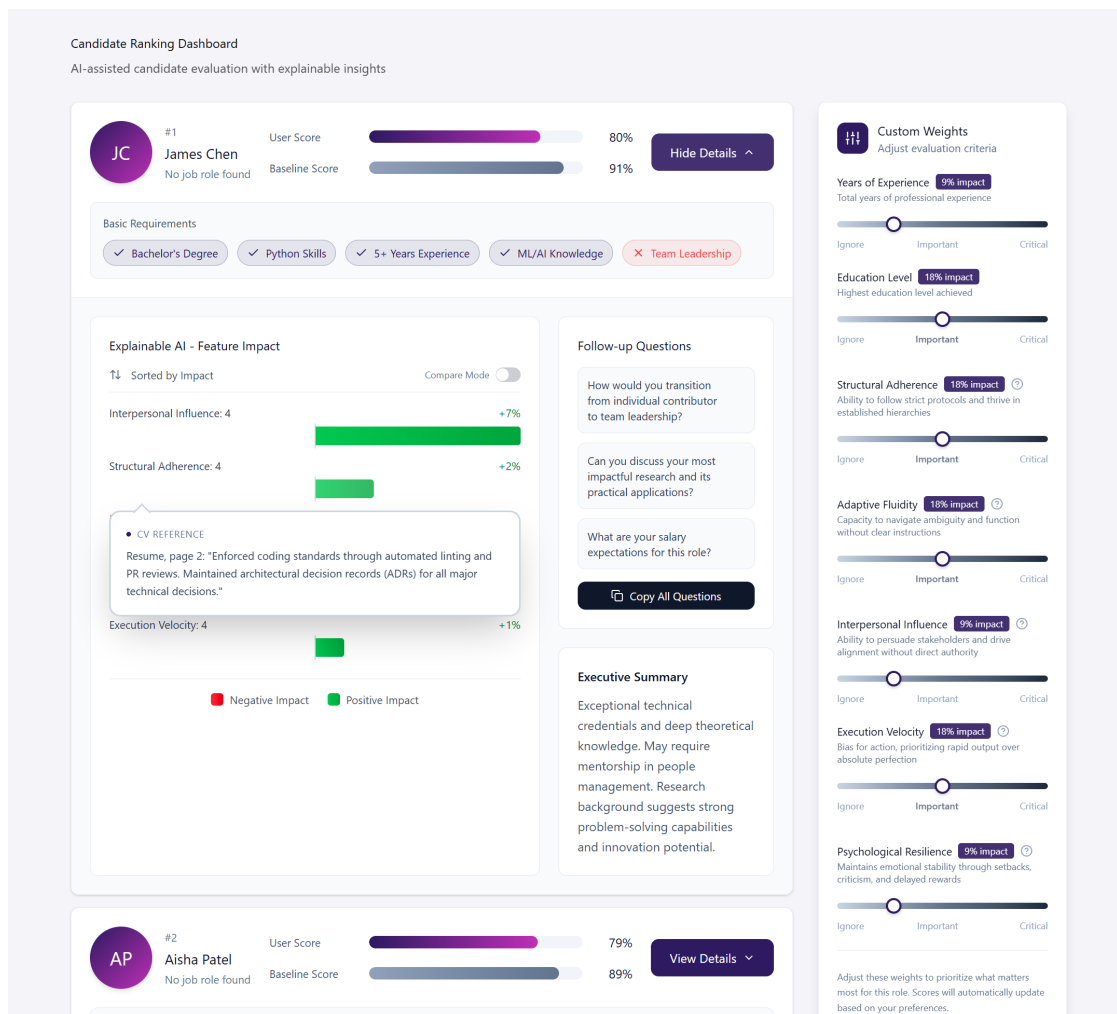
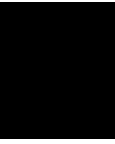


Figure 7.1: Full dashboard view with all implemented improvements



Limitations and Methodological Constraints

While this thesis provides robust evidence for the framework’s success, several methodological constraints must be acknowledged.

First, the evaluation was constrained by the deliberate withholding of the original applicant documents. While the underlying Random Forest model was training on a synthetic, LLM-generated dataset to actively mitigate the historical biases and proxy variables inherent in real-world recruitment data [Fab24, HMO25], the candidate profiles evaluated during the user study were entirely authentic. The job description and the upload samples were sourced from publicly available, real-world CVs of actual software developers [Abr16]. However, access to these documents was intentionally prohibited during the interactive screening task. This methodological restriction was necessary to force participants to engage directly with the generated summaries, TreeSHAP visualizations and dynamic sliders. If participants had been given access to the raw CVs, it could not have been guaranteed that they were evaluating the XAI system itself rather than relying on their conventional manual reading habits. Nevertheless, this artificial constraint triggered the participants’ strong desire for document verification. A real-world deployment evaluation must therefore allow recruiters to interact seamlessly with both the AI dashboard and the original applicant documents to truly measure the system’s impact on operational efficiency.

Second, the evaluation was conducted with a sample size of 14 university Informatics students. While existing literature validates sample sized under 50 for resource, intensive Think-Aloud studies and supports the use of Informatics students as “proxy users” for evaluating technical interfaces [AAATAM24], this demographic inherently lacks the extensive domain expertise of HR professionals. An experienced recruiter might possess a fundamentally different mental model regarding soft skill evaluation and might interact

with the dynamic weighting system using different strategies. Future research must validate these findings through longitudinal field studies within active HR departments to assess the system's long-term impact on hiring diversity and organizational efficiency.

Third, the qualitative data was not evaluated using a standardized metric, but rather derived directly from participant quotes, with key terms and recurring statements systematically synthesized in a summary table.

Conclusion

The integration of AI into recruitment has highlighted an “actionability gap” within the domain of HR, making it hard for recruiters to actively question and verify a recruitment system’s output. Traditional XAI methods, such as SHAP, provide mathematically precise feature contributions, but often remain ambiguous for non-technical users. Providing this explainability without actually improving the insights gained into a system fosters an “illusion of fairness” [Fab25]. This thesis demonstrated that a hybrid architecture balances the computational capabilities of LLMs with a transparent Random Forest scoring mechanism. Specifically, LLMs were used for data extraction and natural language generation of candidate summaries and follow-up interview questions. However, a Random Forest classifier was used for candidate scoring and paired with TreeSHAP to enable a mathematically grounded breakdown of which specific features influenced a candidate’s score. The system’s empirical evaluation showed that users did not blindly trust the system’s recommendations. Instead, users showed analytical skepticism, questioning the system’s outputs using the provided features and exhibiting the need for verification with the original application documents. These findings confirm that AI in recruitment must act as an assistant rather than a fully autonomous replacement.

A contribution of this thesis to new knowledge is the empirical evidence that providing users with dynamic control supports the “bargaining” process. Dynamic control over feature weights gives users a sense of control, allowing them to adapt the AI’s baseline to specific needs. By supporting the Human-in-the-Loop principle by making recruiters actively contribute to the system’s outcomes, automation bias is prevented and “calibrated trust” is nurtured [MB19]. Furthermore, it was demonstrated that providing users with candidate summaries and follow-up interview questions closes the actionability gap, allowing recruiters to question and verify the AI’s outcomes. The generated insights equip recruiters with blueprints to verify the AI’s recommendations in the real world. This demonstrates how generative models can bridge the divide between passive algorithmic outputs and Human-in-the-Loop decision-making.

However, this thesis also identified a challenge in parsing resumes using generative models: the “secondary black box” phenomenon. Using LLMs to infer soft skills from candidate resumes triggers user skepticism if no evidence of the retrieval of these skills is given. This skepticism demonstrates the limits of current LLM transparency.

To address these challenges, a set of recommendations and design guidelines was derived from the evaluation. These guidelines and recommendations are designed to support future generations of actionable XAI tools. First, systems must implement strict data traceability. When using LLMs to infer abstract traits from resumes, a traceback mechanism that directly cites the specific text used for a feature must be implemented. Second, application document access must be maintained. This can be done using side-by-side comparisons, showing the system’s assessment and outputs besides the original application documents. Third, candidate profiles should be integrated with hard skill checkmarks. Representing necessary job requirements as checkmarks allows recruiters to decrease the number of applicants with reduced cognitive load. Additionally, interfaces must design calibration points to calibrate user trust [MB19]. Calibration points are represented as additional visual information that highlight a potential discrepancy with the system’s recommendations. These calibration points must be supported by an explanation as to why the discrepancy was detected. Finally, feature impact diagrams must ensure standardized views to allow direct comparison across all candidates.

This thesis established a functional prototype and validated the proposed framework. However, future research must include longitudinal field studies within active HR departments. A long-term study using a real-world workflow is critical to validate the system’s long term impact and effects on hiring diversity. By embedding the introduced human-centric principles into the design of future XAI applications, future recruitment systems can ultimately evolve into transparent, trustworthy and actionable assistants in talent acquisition.

Raw LLM Outputs

A.1 Prompt Engineering Result for CV Features

The following text is the unedited output generated by Gemini 3 Pro in response to the prompt detailed in section 3.3.4.

Here is a breakdown of “hidden” CV features for the five distinct Software Developer profiles. These soft skills and behavioral traits are tailored to the specific organizational psychology of each environment.

1. Profile: The Pentagon (Defense & High-Security)

- **Context:** A highly hierarchical, bureaucratic, and zero-trust environment where security clearance, protocol, and national security supersede innovation speed or user experience.
- **Standard Feature:** Security Clearance (Secret/Top Secret), C++/Java/Ada proficiency.

Additional Soft-Skill Features:

- **Procedural Rigor:** The ability to adhere strictly to Standard Operating Procedures (SOPs) without deviation. In this environment, “hacking a solution” is a security risk, not a virtue.
- **Compartmentalized Discretion:** A behavioral tendency to keep work context isolated. The candidate must demonstrate the ability to operate effectively without knowing the “full picture” (Need-to-Know basis) and resist the urge to over-share.
- **Hierarchical Deference:** A respect for the chain of command. Unlike tech startups that value “flat structures,” success here depends on navigating and respecting established authority and approval layers.

- **Legacy Systems Patience:** The emotional resilience to work with outdated technology stacks (e.g., COBOL, early Java) without frustration. It reflects a mindset that prioritizes stability and reliability over “shiny object syndrome.”
- **Risk Aversion:** A proactive focus on worst-case scenarios. The ideal candidate prioritizes preventing failure (safety/security) over optimizing for success (speed/features).

2. Profile: R&D Engineer (e.g., Google X, Bell Labs)

- **Context:** A “Moonshot” culture characterized by high ambiguity, long time horizons, deep funding, and a high failure rate. Success is measured by breakthrough discovery, not quarterly shipping.
- **Standard Feature:** PhD or Masters in a specialized field (AI, Quantum, Materials).

Additional Soft-Skill Features:

- **Ambiguity Tolerance:** The ability to remain productive and motivated when the project requirements are undefined and the end goal is theoretically impossible.
- **Intellectual Humility:** The willingness to abandon a beloved hypothesis immediately when data suggests it is wrong. This prevents the “sunk cost fallacy” which drains R&D budgets.
- **Delayed Gratification:** R&D cycles can take years. The candidate must be motivated by the pursuit of knowledge rather than the dopamine hit of daily/weekly deployments.
- **Cross-Disciplinary Curiosity:** A tendency to look outside computer science (e.g., to biology or physics) for solutions. R&D often requires synthesizing disparate fields.
- **Failure Documentation:** A behavioral trait where the candidate meticulously documents why something didn’t work. In R&D, a negative result is a valuable asset if communicated clearly to prevent redundancy.

3. Profile: Consultant (e.g., McKinsey Digital, BCG Platinion)

- **Context:** High-pressure, client-facing, travel-heavy (virtual or physical), and project-based. Developers must sell their expertise as much as they code.
- **Standard Feature:** Full-stack capability, agility with modern frameworks.

Additional Soft-Skill Features:

- **Executive Presence:** The ability to communicate technical debt or architectural risks to non-technical C-suite stakeholders in terms of P&L (Profit and Loss) and business risk.
- **Context Switching Agility:** The behavioral flexibility to drop a project on Friday and be fully operational on a completely different tech stack and industry vertical by Monday morning.
- **Diplomatic Assertiveness:** The skill to tell a client their current architecture is failing without causing offense. It requires balancing “the customer is right” with “the customer needs to change.”
- **80/20 Pragmatism:** A focus on delivering the “Minimum Viable Product” that solves the business case, rather than pursuing code perfection. Consultants must prioritize speed-to-value over academic purity.
- **Social Chameleonism:** The ability to mirror the culture of the client (e.g., wearing a suit at a bank vs. a hoodie at a media firm) to build rapid trust and rapport.

4. Profile: Early-Stage Start-up (Seed to Series A)

- **Context:** Chaos, existential threat (running out of cash), role fluidity, and a “move fast and break things” mentality.
- **Standard Feature:** History of hackathons, side projects, or previous founder experience.

Additional Soft-Skill Features:

- **Ruthless Prioritization:** The behavioral trait of ignoring 90 % of “good ideas” to focus on the one feature that ensures survival/revenue this month.
- **Role Fluidity (Low Ego):** The willingness to jump from backend coding to customer support, data entry, or ordering pizza. “That’s not my job” is a disqualifying phrase here.
- **Bias for Action:** A preference for making a reversible wrong decision quickly over waiting for a perfect decision. Analysis paralysis kills startups.
- **Thrift/Resourcefulness:** A hacker mentality that looks for the cheapest, fastest solution (e.g., stringing together APIs) rather than building custom enterprise-grade solutions from scratch.
- **Optimistic Resilience:** The emotional stability to handle the “roller-coaster”—high highs of a new sale and low lows of a system crash—without burning out or losing faith in the vision.

5. Profile: Open-Source Maintainer (e.g., Linux Kernel, React)

- **Context:** Asynchronous, public, community-driven, and often volunteer-based. Authority is earned through contribution and consensus, not granted by title.
- **Standard Feature:** Significant GitHub contribution graph, history of pull requests.

Additional Soft-Skill Features:

- **Asynchronous Clarity:** The ability to write exhaustive, unambiguous documentation and communication. Because there are no “meetings,” the text must stand alone and guide contributors across time zones.
- **Consensus Building:** The soft skill of steering a community of volunteers toward a technical decision without having direct managerial authority over them.
- **Emotional Detachment (Thick Skin):** The ability to separate one’s self-worth from one’s code. Public code review can be brutal; successful maintainers accept critique objectively without taking offense.
- **Mentorship Mindset:** A behavioral tendency to nurture new contributors. A maintainer cannot scale if they are the only one who understands the codebase; they must actively teach others.
- **Stewardship (Long-Term Thinking):** Unlike startups, this profile requires extreme caution regarding breaking changes. It values backward compatibility and stability for the ecosystem over “new features.”

Next Step

Would you like me to help you map these behavioral traits to specific **keywords or semantic patterns** that your XAI model could look for in CV text or cover letters?

User Study Materials

B.1 Question Survey

The question survey used for quantitative evaluation can be found below. There are a total of 38 questions, which can each be answered using a standard 5-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). It is divided into five logical parts:

1. Explanation Satisfaction: measures the clarity, completeness and utility of the TreeSHAP visualizations and LLM summaries
2. Trust in Automation: The user's confidence in the system's underlying Random Forest scoring mechanism
3. Perceived Usefulness (TAM): Whether the XAI features actually helped close the actionability gap and supported their decision-making
4. Cognitive Load: the mental effort required to process the recruiter dashboard and generated narratives.
5. System Usability Score (SUS): General interface friction, navigation and usability of the sliders and layout.

The following questions are the exact questions asked to study participants during the user study:

B.1.1 Part 1: Explanation Satisfaction

1. From the explanation, I understand how the AI model works.
2. The explanation of how the model predicts is satisfying.
3. The explanation of how the model works has sufficient detail.
4. The explanation of how the model works seems complete.
5. The explanation of how the model works tells me how to use it.
6. The explanation of how the model works is useful to my goal.
7. The explanation of the model shows me how accurate the model is.
8. The explanation lets me judge when I should trust and not trust the model.

B.1.2 Part 2: Trust in Automation

9. I believe the model is a competent performer.
10. I trust the model.
11. I have confidence in the prediction provided by the AI model.
12. I can depend on the model to analyse the candidate profiles.
13. I can rely on the model to behave in consistent ways.
14. I can rely on the model to do its best every time I take its advice.

B.1.3 Part 3: Perceived Usefulness (TAM)

15. The AI-powered model enriched the candidate evaluation process.
16. The model was helpful to me in acquiring new insights about the candidates.
17. The features provided by the AI model smoothed the screening process.
18. The model helped me obtain useful information when needed.
19. The AI-powered model helped me evaluate the candidates better.
20. The AI-embedded screening approach is more useful than conventional manual CV screening.

B.1.4 Part 4: Cognitive Load

21. The information presented in this activity was difficult for me.
22. I had to put a lot of effort into evaluating the candidates in this activity.
23. It was troublesome for me to evaluate the candidates in this activity.
24. I felt frustrated evaluating the candidates in this activity.
25. I did not have enough time to evaluate the candidates in this activity.
26. During the activity, the workflow or information presentation caused me a lot of mental effort.
27. I need to put lots of effort into completing the tasks or achieving the objectives in this screening activity.
28. The workflow in the evaluation task was difficult to follow and understand.

B.1.5 Part 5: System Usability Scale (SUS)

29. I think that I would like to use this interface frequently.
30. I found the interface unnecessarily complex.
31. I thought the interface was easy to use.
32. I think I would need the support of a technical person to use this interface.
33. I found the various functions in this interface were well integrated.
34. I thought there was too much inconsistency in this interface.
35. I would imagine that most people would learn to use this interface quickly.
36. I found the interface very unmanageable (irritating or tiresome) to use.
37. I felt very confident using the interface.
38. I needed to learn a lot of things before I could get going with this interface.

B.2 Semi-Structured Interview Questions

The following semi-structured interview guide for the qualitative interview at the end of each time slot is listed below. Its main objective is to gather qualitative data to contextualize the quantitative findings from the survey. While standard questions were used for all interviews, certain follow-up questions were asked to clarify specific behavior observed during the Think-Aloud protocol.

B.2.1 Theme 1: The “Bargaining” Process

- Did you already have a specific slider setting in mind when interacting with the sliders or did you play around with the sliders a bit to see how the system works?
- Did the ability to manually change the weights of different skills make you feel more or less in control of the recommendation? Why?
- Do you feel like you’d need to see the actual candidate CVs to be able to properly judge them?
- Situational follow-up: How important were the sliders’ presets to you?

B.2.2 Theme 2: Trust and Transparency

- When looking at the horizontal bar charts representing the AI’s reasoning, how did that visualization influence your trust in the candidate’s overall score?
- You encountered an information badge for one of the candidates. How did this automated warning impact your judgment?
- How difficult was it for you to understand the soft skills like “Adaptive Fluidity”? Did the slider descriptions help you understand them better?
- Situational follow-up: Was there any part of the AI’s mathematical explanation that felt confusing or overwhelming to interpret?

B.2.3 Theme 3: Closing the Actionability Gap

- Compared to simply reading a standardized CV, how did reading an additional summary help you understand the candidate’s skills?
- Looking at the generated follow-up interview questions, do you feel those were topics you would have eventually thought to ask yourself, or did the AI help you identify new angles?
- Situational follow-up: If you were officially hiring for this start-up tomorrow, would you use these exact questions?

B.2.4 Closing Feedback

- If you could change anything about this interface to make it more helpful for you as a recruiter, what would it be?

B.3 Transcription of Explanation Video

To guarantee that each participant received the same level of information, a pre-recorded video was played before users could start the task. The transcription of this video can be found below.

Your first step will be to upload the candidate CVs and the job listing provided to the system to initialize the dashboard. For each candidate, you will see two scores. The bottom score is the system’s objective baseline score. The top score is your dynamically weighted score. On the right, you have different sliders for different soft skills and requirements. You can adjust these based on what you think is important for a start-up and the candidates’ rankings will update in real-time. By default, the sliders are already pre-adjusted based on your specific job description. If you expand a candidate’s profile, you will see an AI-generated summary, a visual breakdown of why the AI scored them that way, and some suggested interview questions. If the system notices a discrepancy between your weighted score and its baseline score, it might flag a candidate to show you where this discrepancy comes from.

B.4 Question Survey Results

Table B.1: Raw Participant Responses: Explanation Satisfaction

Item	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12	P13	P14
From the explanation, I understand how the AI model works.	4	4	4	4	3	5	5	3	3	3	5	4	3	5
The explanation of how the model predicts is satisfying.	4	3	4	3	2	5	5	5	2	2	4	4	5	5
The explanation of how the model works has sufficient detail.	5	2	3	3	3	4	5	4	4	2	5	3	5	5
The explanation of how the model works seems complete.	5	2	4	3	4	4	5	4	2	3	5	4	4	5
The explanation of how the model works tells me how to use it.	3	4	5	5	5	5	5	5	5	3	5	5	5	5
The explanation of how the model works is useful to my goal.	4	4	4	4	4	5	5	5	4	3	4	5	5	5
The explanation of the model shows me how accurate the model is.	1	3	5	3	2	4	5	3	3	2	4	4	4	4
The explanation lets me judge when I should trust and not trust the model.	2	3	5	2	3	3	4	4	5	2	3	3	4	3

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Table B.2: Raw Participant Responses: Trust in Automation

Item	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12	P13	P14
I believe the model is a competent performer.	3	4	5	3	3	4	5	4	1	2	3	2	4	2
I trust the model.	3	2	3	3	2	4	5	4	3	3	2	2	4	2
I have confidence in the prediction provided by the AI model.	4	2	4	3	3	4	5	5	3	4	4	3	4	2
I can depend on the model to analyse the candidate profiles.	3	3	3	3	4	4	5	4	5	3	4	3	4	4
I can rely on the model to behave in consistent ways.	4	4	3	2	4	4	4	2	3	3	5	5	2	5
I can rely on the model to do its best every time I take its advice.	4	4	3	3	3	5	4	2	2	4	3	5	3	5

Table B.3: Raw Participant Responses: Perceived Usefulness

Item	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12	P13	P14
The AI-powered model enriched the candidate evaluation process.	4	3	5	4	4	3	4	5	4	5	4	5	4	2
The model was helpful to me in acquiring new insights about the candidates.	4	4	4	4	4	4	4	5	4	5	4	5	5	3
The features provided by the AI model smoothed the screening process.	5	2	5	4	3	4	5	4	4	5	5	5	4	2
The model helped me obtain useful information when needed.	4	3	4	3	3	5	5	4	4	3	5	4	4	4
The AI-powered model helped me evaluate the candidates better.	4	3	4	4	3	4	5	5	2	5	4	4	3	3
The AI-embedded screening approach is more useful than conventional manual CV screening.	3	1	4	5	3	3	3	4	4	4	2	3	4	1

Table B.4: Raw Participant Responses: Cognitive Load

Item	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12	P13	P14
The information presented in this activity was difficult for me.	1	3	1	2	1	1	1	3	4	1	1	2	1	1
I had to put a lot of effort into evaluating the candidates in this activity.	2	5	2	2	2	2	1	2	4	1	1	3	2	1
It was troublesome for me to evaluate the candidates in this activity.	1	4	1	3	4	1	1	2	4	2	1	2	1	1
I felt frustrated evaluating the candidates in this activity.	1	4	1	1	4	1	1	1	4	1	1	1	2	1
I did not have enough time to evaluate the candidates in this activity.	1	5	3	1	1	1	1	1	5	1	1	1	1	1
During the activity, the workflow or information presentation caused me a lot of mental effort.	1	2	1	2	1	1	1	1	3	1	1	2	3	1
I need to put lots of effort into completing the tasks or achieving the objectives in this screening activity.	1	4	2	3	1	1	1	1	5	1	1	2	2	1
The workflow in the evaluation task was difficult to follow and understand.	1	2	1	1	1	1	1	1	3	1	1	2	1	1

Table B.5: Raw Participant Responses: System Usability Scale (SUS)

Item	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12	P13	P14
I think that I would like to use this interface frequently.	4	3	5	4	3	4	4	4	3	5	3	4	4	2
I found the interface unnecessarily complex.	1	1	1	1	1	1	1	1	4	2	1	2	2	1
I thought the interface was easy to use.	4	4	5	5	5	5	5	5	3	5	5	4	5	5
I think I would need the support of a technical person to use this interface.	1	1	1	1	1	1	1	1	1	1	1	1	1	1
I found the various functions in this interface were well integrated.	4	4	4	4	4	4	5	5	5	5	5	4	4	5
I thought there was too much inconsistency in this interface.	1	1	1	3	1	1	1	1	1	1	1	4	1	1
I would imagine that most people would learn to use this interface quickly.	5	4	5	5	5	5	5	5	4	5	5	5	5	5
I found the interface very unmanageable (irritating or tiresome) to use.	2	1	1	1	1	1	1	1	3	1	1	1	1	1
I felt very confident using the interface.	4	3	4	4	4	5	5	4	2	4	5	5	4	5
I needed to learn a lot of things before I could get going with this interface.	2	1	1	1	1	1	1	1	3	1	1	1	1	1

Overview of Generative AI Tools Used

To support the research and writing process of this thesis, Google Gemini was employed as a developmental and editorial assistant. Its specific applications were:

- Technical Implementation: Facilitating the development process by assisting with code generation, structural recommendation and debugging for the application prototype
- Document Formatting: Assisting with LaTeX typesetting, specifically by resolving compilation errors and refining the layout of tables and figures
- Language Improvement: Serving as a tool for language refinement to improve clarify and tone

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